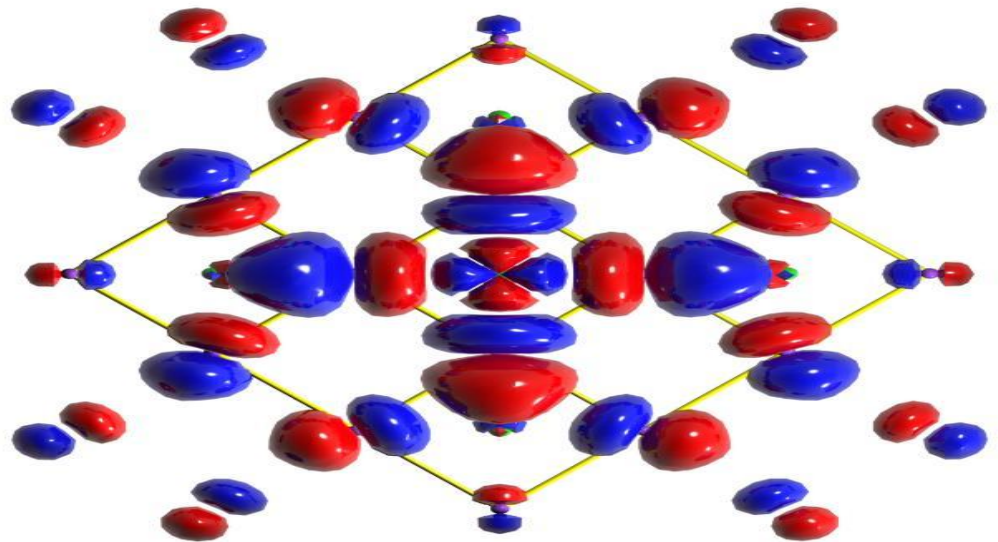


Quantum Theory and the Electronic Structure of Atoms

Chapter 1B



Chapter 1B outline

1. Properties of waves
2. Planck's Quantum Theory
3. Bohr's Atomic Model
4. Rydberg Equation
5. Emission spectra
6. Formation of continuous and line spectrum
7. Atomic hydrogen emission in various series
8. Quantum mechanical model
9. Quantum numbers
10. Shape of atomic orbitals
11. Electronic configuration

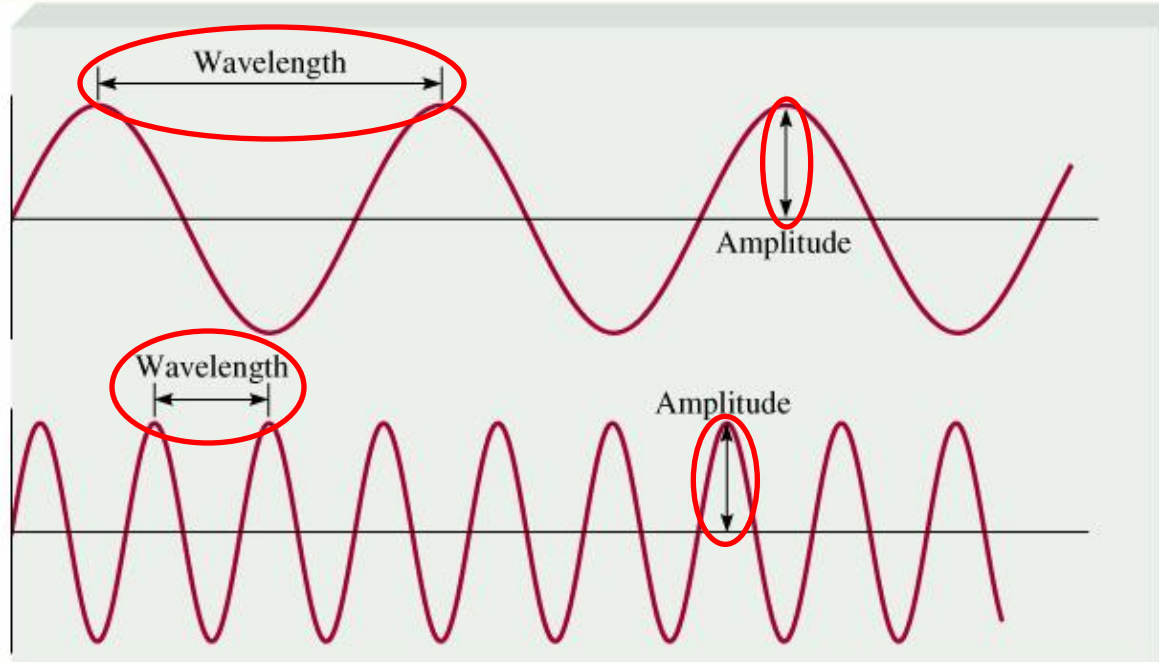
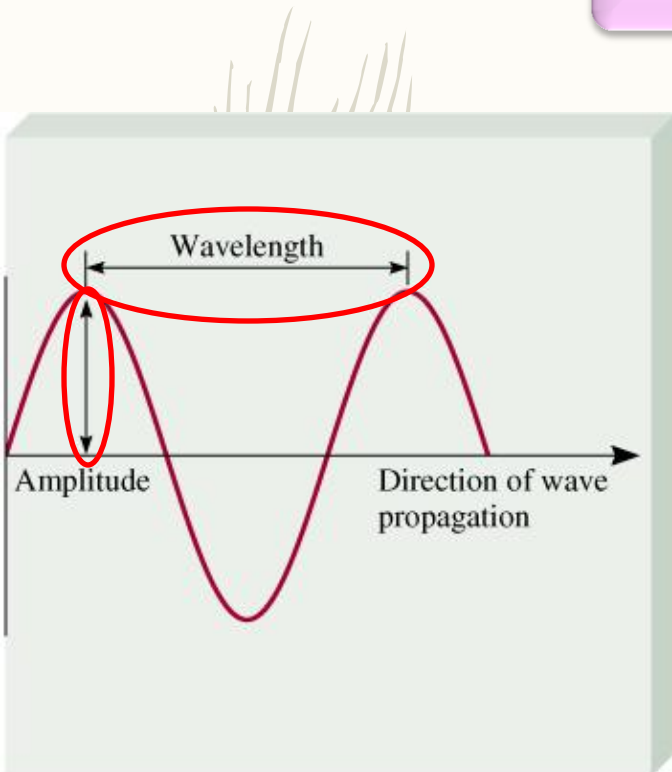
LEARNING OBJECTIVES

- Summarize the properties of waves including wavelength, frequency and amplitude.
- Interconvert wavelength and frequency of knowing the speed of light constant.
- Evaluate the basis of quantum theory according to Planck's work.
- Use the Planck equation to determine wavelength, frequency and energy of photon.
- Generalize the Bohr atomic model to explain atomic emission spectra in hydrogen.
- Categorize the emission spectra into continuous and line (atomic) spectrum
- Relate the energy levels in the hydrogen atom with various emission series.

LEARNING OBJECTIVES (continue..)

- Understand the meaning of the four quantum numbers required to describe an electron in an atom.
- Adopt quantum number rules to determine allowed values for each quantum number.
- Distinguish the basis of atomic orbitals and theirs corresponding to quantum numbers.
- Compare the arrangement of atomic orbitals by energy levels in multi-electron atoms.
- Adoption of the Aufbau principle, Pauli exclusion principle and Hund's rule in determining electron configurations and drawing electron orbitals diagram.
- Demonstrate the use of periodic table to write electron configurations of atoms.
- Identify anomalous electronic configuration of certain elements.
- Recognize the electron configuration of ions.

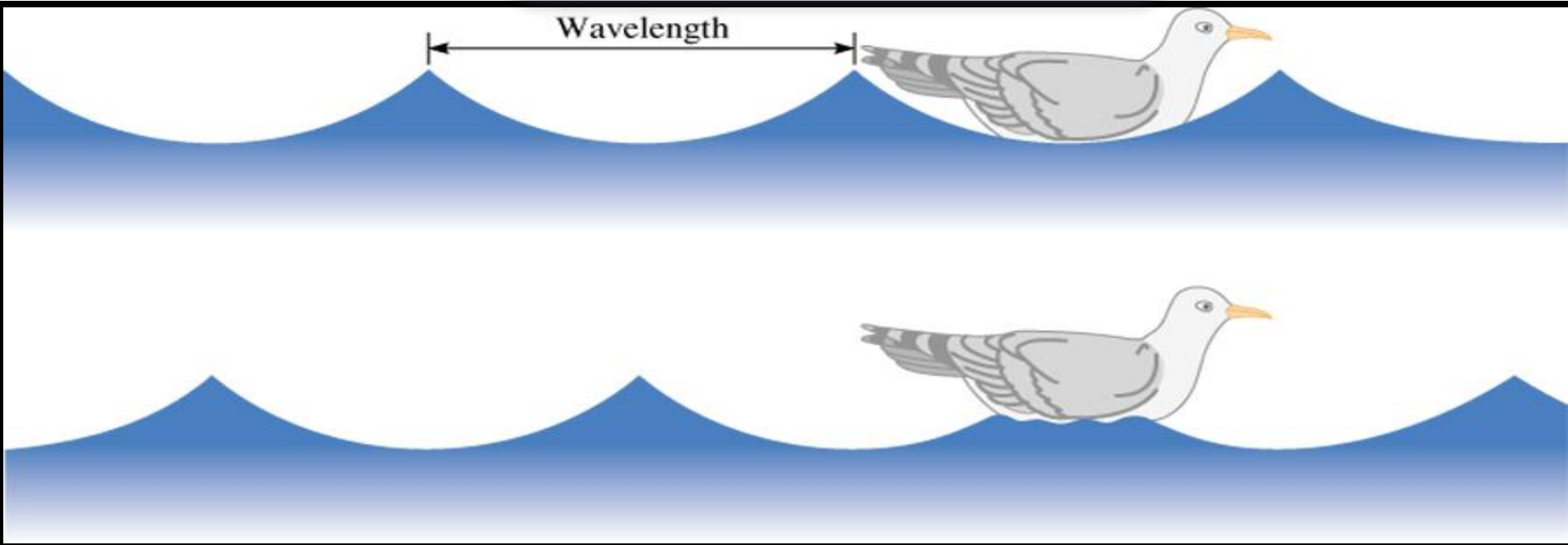
Properties of Waves



Wavelength is the distance between identical points on successive waves.

Amplitude is the vertical distance from the midline of a wave to the peak or trough.

Properties of Waves



Frequency (n) is the number of waves that pass through a particular point in 1 second ($\text{Hz} = 1 \text{ cycle/s}$).

Speed depends on the type of wave and nature of the medium through which the wave is travelling.

Planck's Quantum Theory

Energy (light) is emitted or absorbed in discrete units (quantum).

$$E = h \times \nu$$

Planck's constant , $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$

Speed (u) of the wave = wavelength x frequency
 $= \lambda \times \nu$

Speed of light (c) in vacuum = 3.00×10^8 m/s

All electromagnetic radiation
 $\lambda \times \nu = c$

Try this...

a. Calculate the speed of a wave whose wavelength and frequency are 17.4 cm and 87.4 Hz respectively.

b. The wavelength of the green light from a traffic signal is 522 nm. What is the frequency of this radiation?

The formula relating the frequency f and wavelength λ is

$$f\lambda = c \text{ or } f = \frac{c}{\lambda}$$

where c is the speed of light.

In your case, $\lambda = 522 \text{ nm} = 522 \times 10^{-9} \text{ m}$ and $c = 2.998 \times 10^8 \text{ m}\cdot\text{s}^{-1}$.

$$\therefore f = \frac{c}{\lambda} = \frac{2.998 \times 10^8 \text{ m}\cdot\text{s}^{-1}}{522 \times 10^{-9} \text{ m}} = 5.74 \times 10^{14} \text{ s}^{-1}$$





Try this...

$$E = h \times \nu$$

Planck's constant , $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$

Calculate the energy (in joules) of

a) a photon with a wavelength of $5.00 \times 10^4 \text{ nm}$ (infrared region)

b) a photon with a wavelength of $5.00 \times 10^{-2} \text{ nm}$ (X ray region)

$$E = h \times \nu$$

$$E = h \times c / \lambda$$

$$E = 6.63 \times 10^{-34} (\text{J}\cdot\cancel{\text{s}}) \times 3.00 \times 10^8 (\cancel{\text{m/s}}) / 50000 \times 10^{-9} (\cancel{\text{m}})$$

$$E = \underline{\hspace{2cm}} \text{ J}$$

2) What is the wavelength (in m) of an electromagnetic wave whose frequency is $3.64 \times 10^7 \text{ Hz}$.



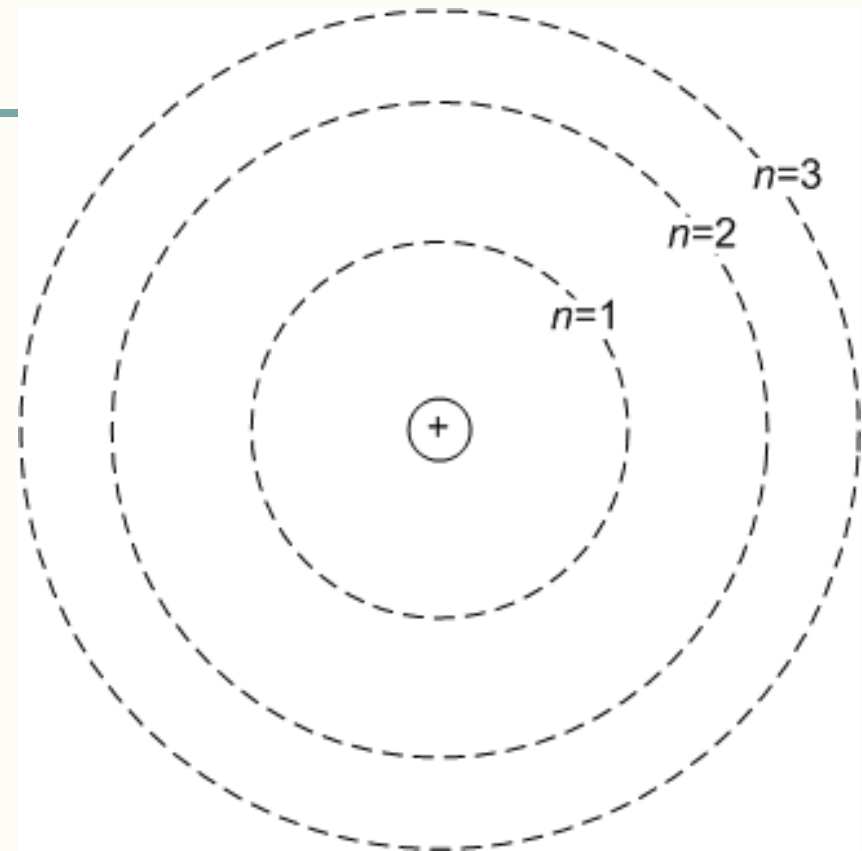
Bohr's Atomic Model

- In 1913, a young Dutch physicist, Niels Böhr proposed a theory of atom that shook the scientific world.
- The atomic model he described had **electrons circling a central nucleus** that contains positively charge proton.



Bohr's Atomic Model

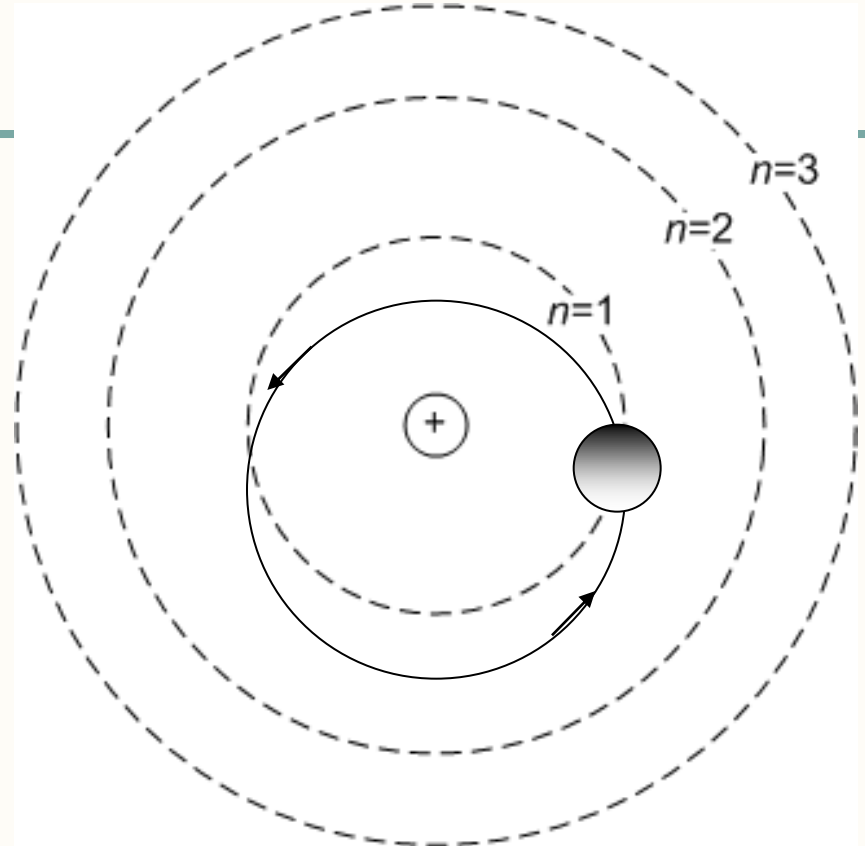
- Bohr also proposed that these orbits can only occur at **specifically “permitted” levels** according to the energy levels of the electron and explain successfully the lines in the hydrogen spectrum.



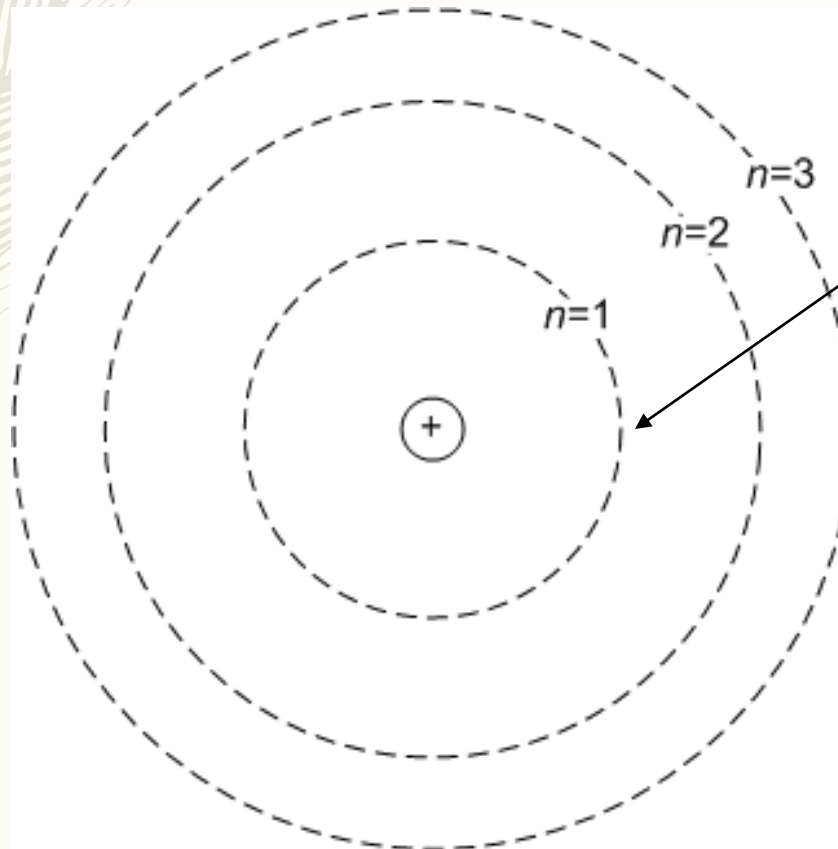
Bohr's Atomic Model

First Postulates :

- Electron moves in circular orbits around the nucleus.
- While moving in the orbit, the electron does not radiate or absorb any energy.



- [orbit = energy level=shell]
- Orbit is a pathway where the electron is move around the nucleus.

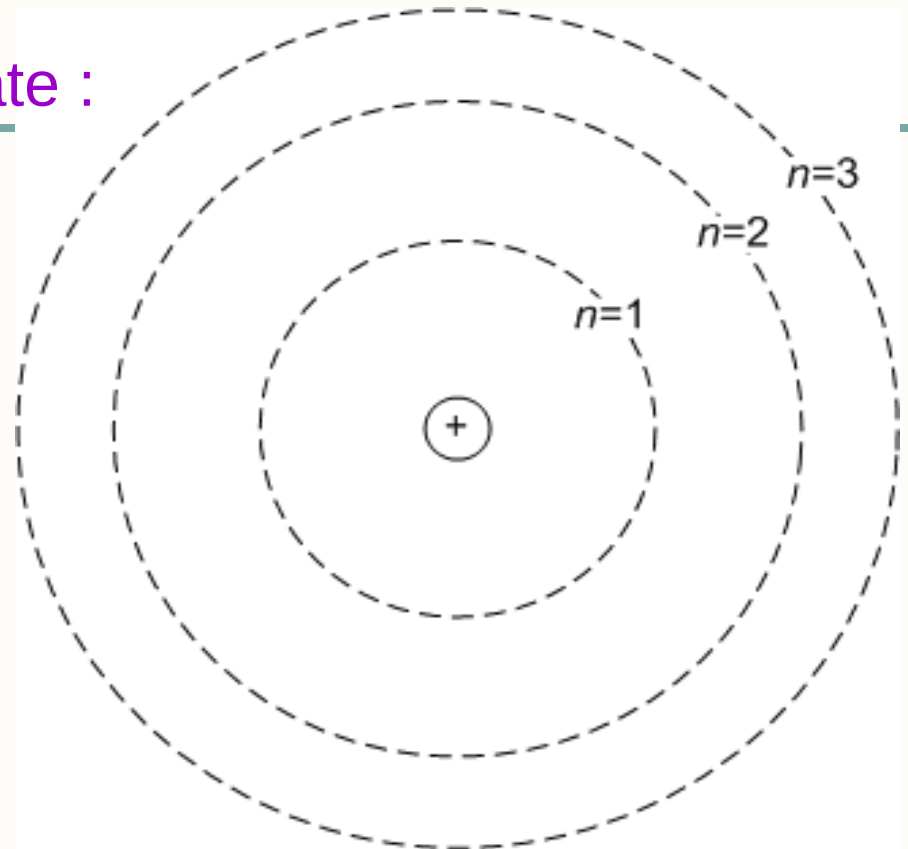


Orbit

Bohr's Atomic Model

Second Postulate :

- The moving electron has a specific amount of energy; its energy is quantized.



The energy of an electron in its level is given by:

$$E_n = - R_H \frac{1}{n^2}$$

R_H (Rydberg constant) = $2.18 \times 10^{-18} \text{J}$.

n (principal quantum number) = 1, 2, 3 ∞ (integer)

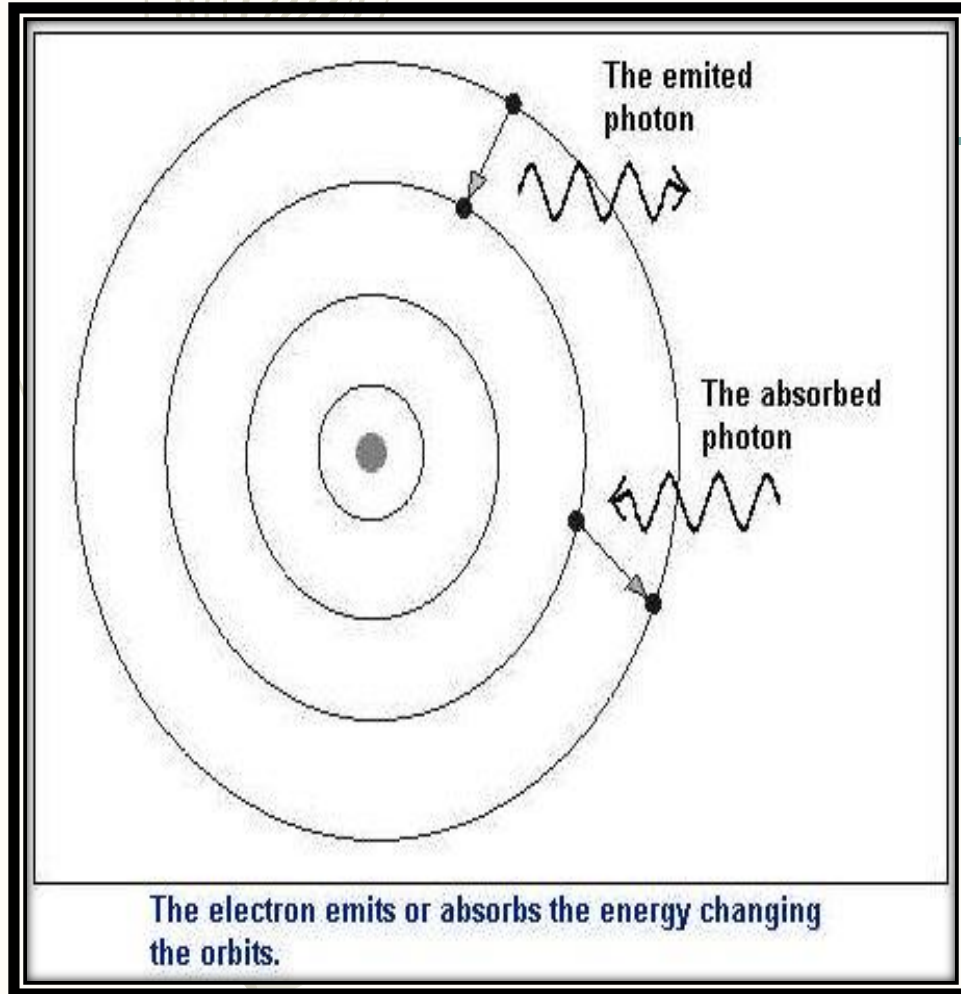
Note:

- n identifies the orbit of electron
- Energy is zero if electron is located infinitely far from nucleus

Bohr's Atomic Model

Third postulate :

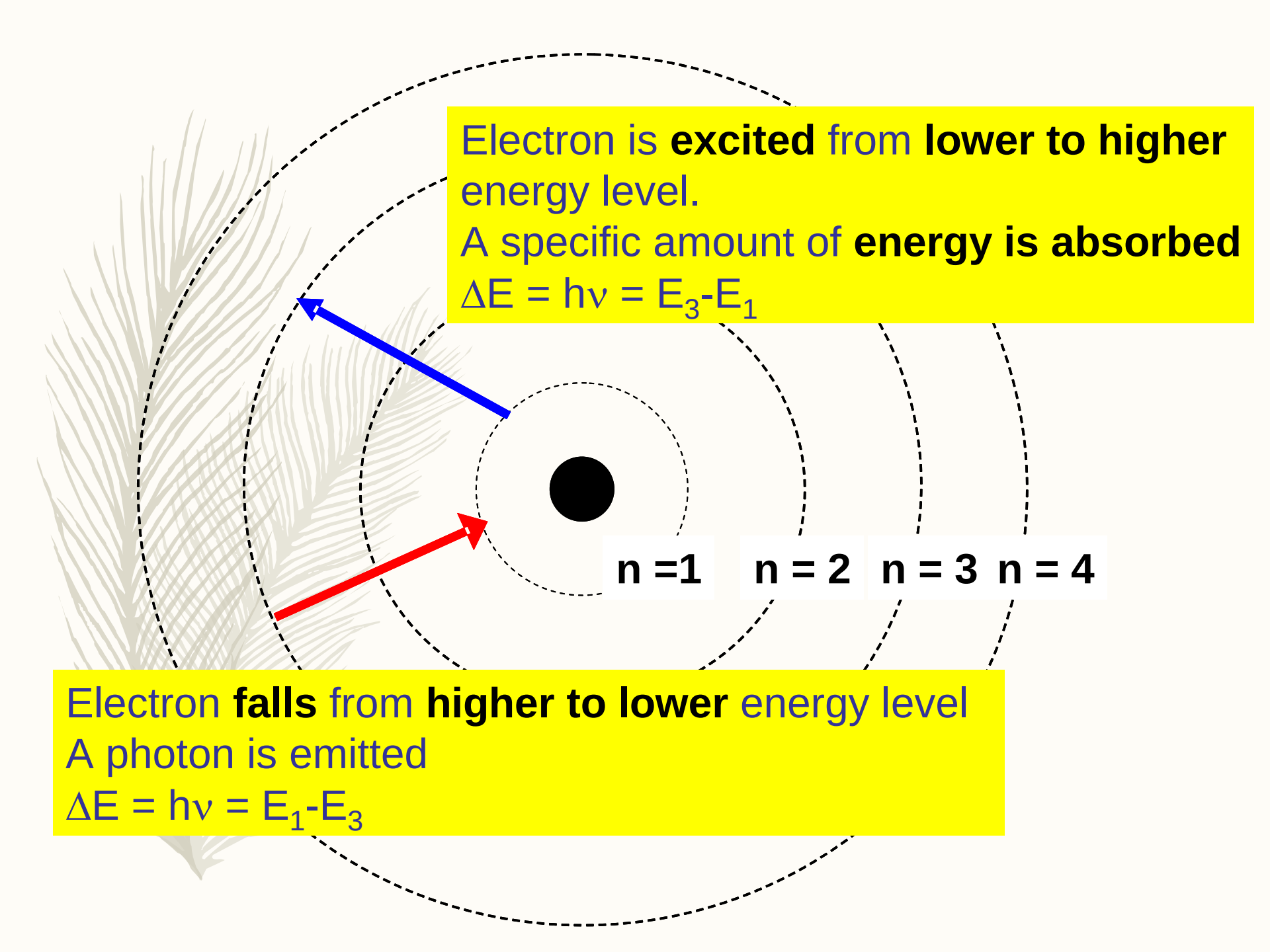
- At ordinary conditions, the electron is at the ground state (lowest level).
- If energy is supplied, electron absorbed the energy and is promoted from a lower energy level to a higher ones. (Electron is excited)



Bohr's Atomic Model

Fourth Postulate :

- Electron at its excited states is unstable.
- It will fall back to lower energy level and released a specific amount of energy in the form of light (photon).
- The energy of the photon equals the energy difference between levels.



The diagram shows a central black nucleus with four concentric dashed circles representing energy levels. The levels are labeled n=1, n=2, n=3, and n=4 from innermost to outermost. A blue arrow points from the n=1 level to the n=3 level, representing an excitation. A red arrow points from the n=3 level to the n=1 level, representing a fall. A faint background image of a feather is visible on the left side.

Electron is **excited** from **lower** to **higher** energy level.

A specific amount of **energy** is absorbed
 $\Delta E = h\nu = E_3 - E_1$

n = 1

n = 2

n = 3

n = 4

Electron **falls** from **higher** to **lower** energy level

A photon is emitted

$$\Delta E = h\nu = E_3 - E_1$$

- **Radiant energy emitted when the electron moves from higher-energy state to lower-energy state is given by:**

$$\Delta E = E_f - E_i$$

$$E_f = -R_H \frac{1}{n_f^2}$$

$$E_i = -R_H \frac{1}{n_i^2}$$

$$\Delta E = -R_H \left(\frac{1}{n_f^2} \right) - \left(-R_H \left(\frac{1}{n_i^2} \right) \right)$$

$$\text{Thus, } \Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

n (principal quantum number) = 1,2,3,...

R_H (Rydberg constant) = $2.18 \times 10^{-18} \text{J}$

Rydberg Equation

Wavelength emitted by the transition of electron between two energy levels is calculated using Rydberg equation:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

n (principal quantum number) = 1,2,3,...

R_H (Rydberg constant) = $1.09677 \times 10^7 \text{ m}^{-1}$

Bohr's Model of the Atom (1913)

1. e^- can only have specific (quantized) energy values
2. light is emitted as e^- moves from one energy level to a lower energy level

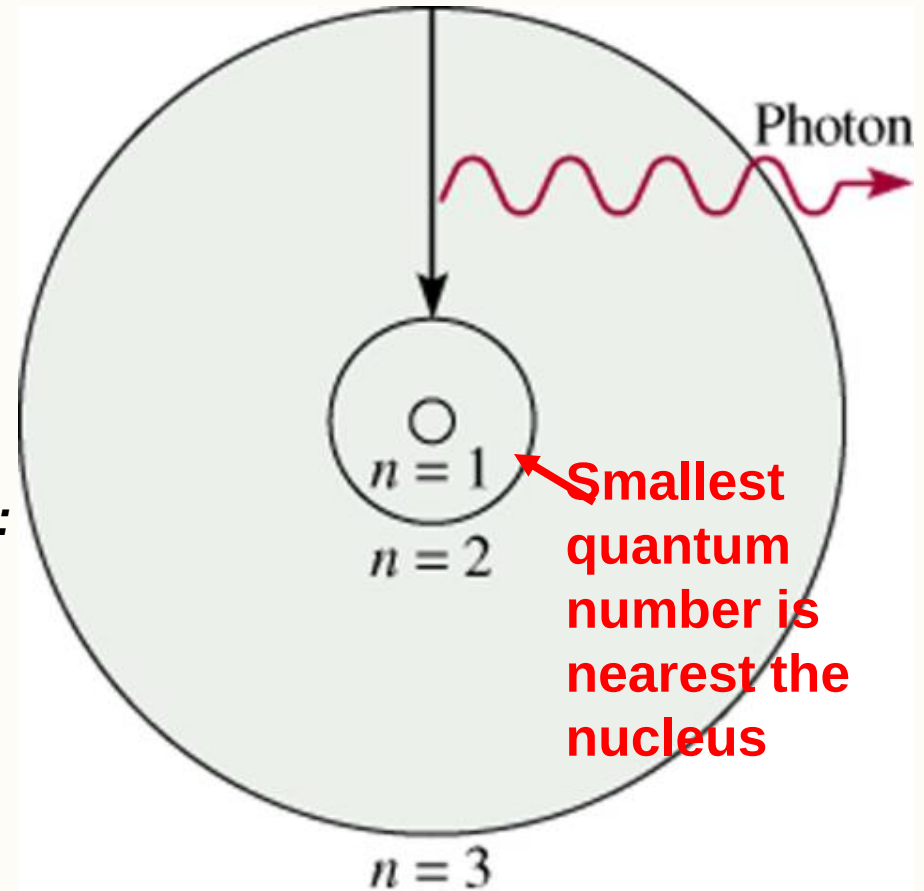
Energy for an electron in hydrogen atom:

$$E_n = -R_H \left(\frac{1}{n^2} \right)$$

n (principal quantum number) = 1, 2, 3, ...

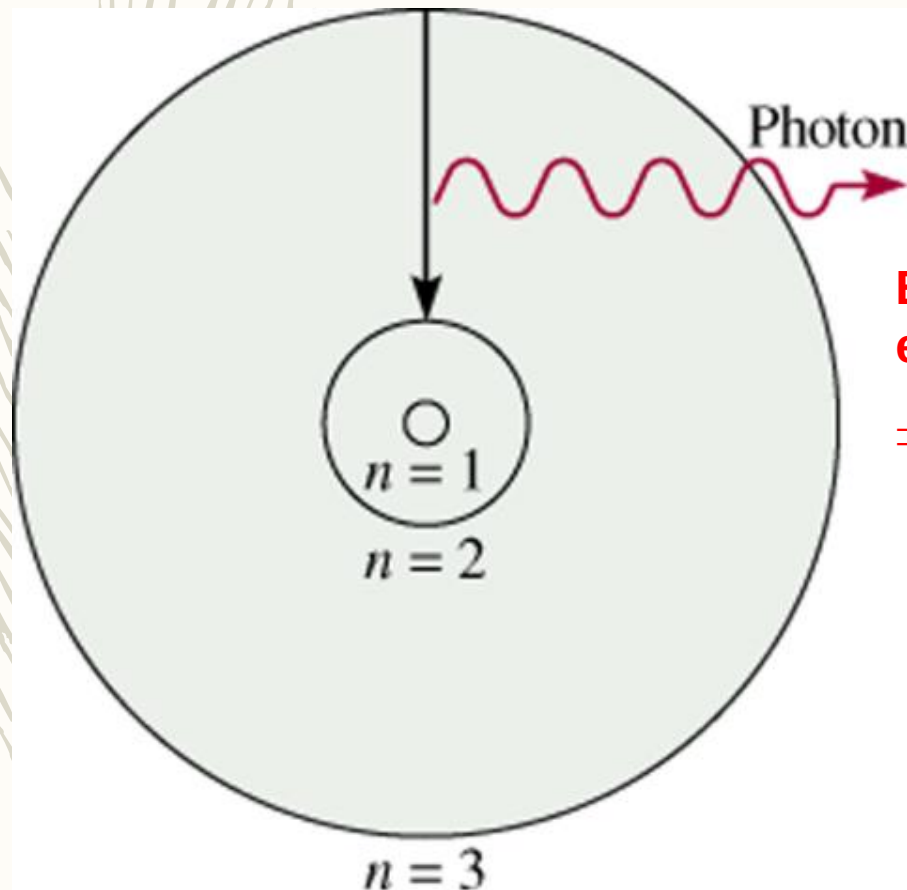
R_H (Rydberg constant) = $2.18 \times 10^{-18} \text{ J}$

Model takes ideas from astronomy – planets moving around the sun => electrons in orbit around the nucleus



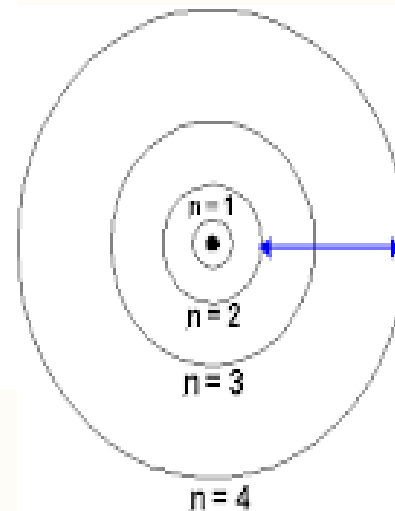
Negative sign signifies that energy is lower than a "free electron"

Bohr's Model of the Atom Has Electrons Moving in Circular Orbits



Each orbit has a particular energy

⇒ Energy is quantized



Electron initially in orbit $n = 2$, may absorb just the right energy from a photon and jump to $n = 4$.

It will quickly drop back to its original orbit emitting a photon of energy equal to the one it originally absorbed.

In both cases the change in energy of the electron is equal to Planck's constant times the frequency of the absorbed or emitted radiation ($\Delta E = hf$).

Bohr's Model of the Atom

$$E_n = -R_H \left(\frac{1}{n^2} \right)$$

n (principal quantum number) = 1,2,3,...

$n = 1$ has the most negative value

⇒ Ground state

⇒ *Ground level*

⇒ *Lowest energy state*

$n = 2$

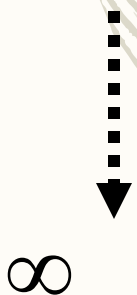
$= 3$

$= 4$

⇒ Higher energy state

⇒ *Excited level*

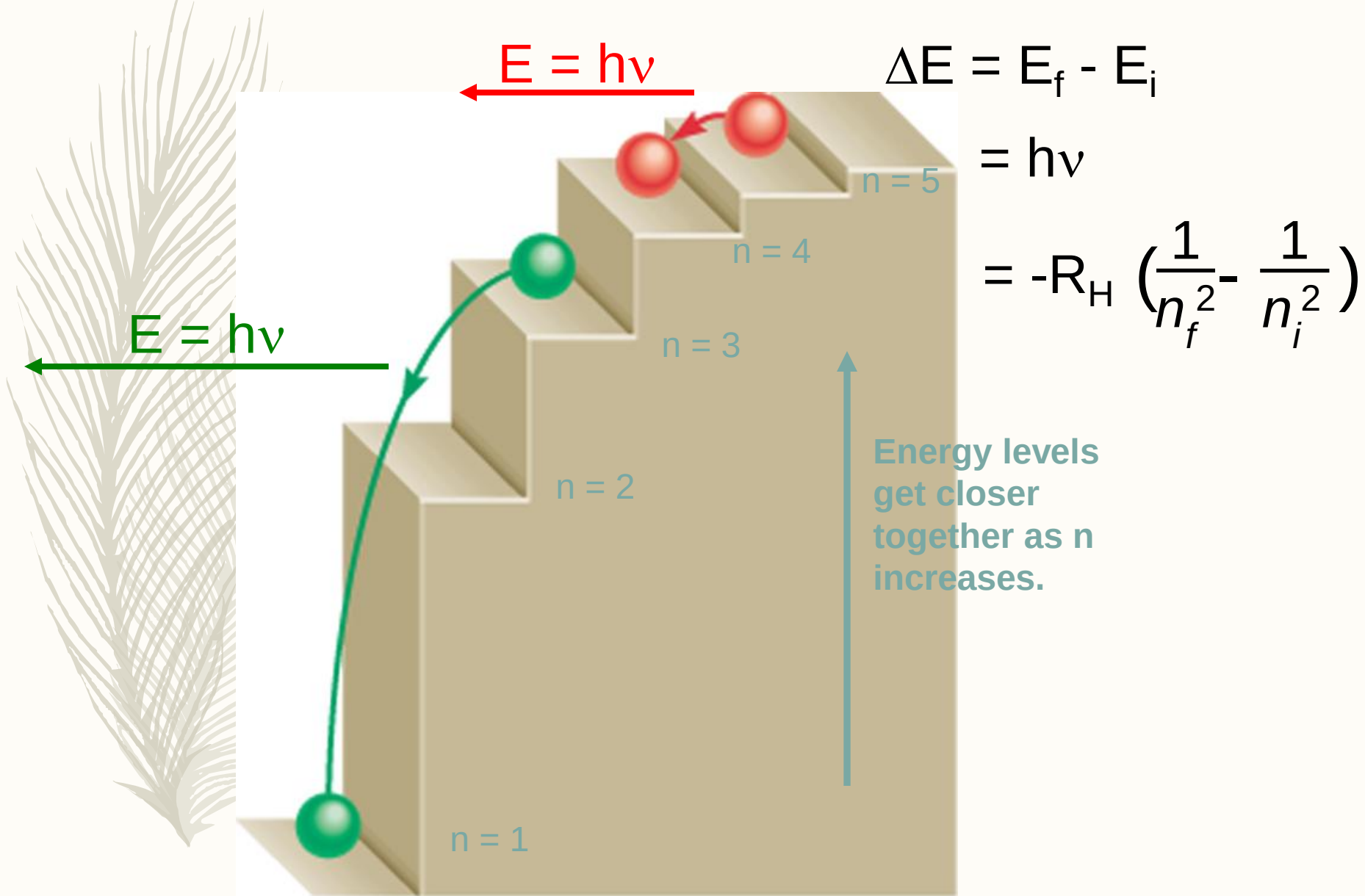
⇒ *Excited state*



$$E_{\infty} = -R_H \left(\frac{1}{\infty^2} \right) \xrightarrow{0} \text{"free electron"}$$

Energy can be emitted or absorbed

⇒ Must go between energy levels

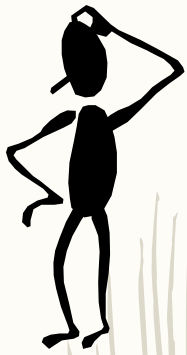


⇒ Must go between energy levels

$E = h\nu$

$E = h\nu$

$$\begin{aligned}\Delta E &= E_1 - E_3 \\ &= h\nu \\ &= -R_H \left(\frac{1}{1^2} - \frac{1}{3^2} \right) \\ &= -0.888 R_H \\ &= -1.94 \times 10^{-18} \text{ J} \\ &\text{emit light}\end{aligned}$$



Calculate the wavelength (in nm) of a photon emitted by a hydrogen atom when its electron drops from the $n = 5$ state to the $n = 3$ state.

$$E_{\text{photon}} = \Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$E_{\text{photon}} = 2.18 \times 10^{-18} \text{ J} \times (1/25 - 1/9)$$

$$E_{\text{photon}} = \Delta E = -1.55 \times 10^{-19} \text{ J}$$

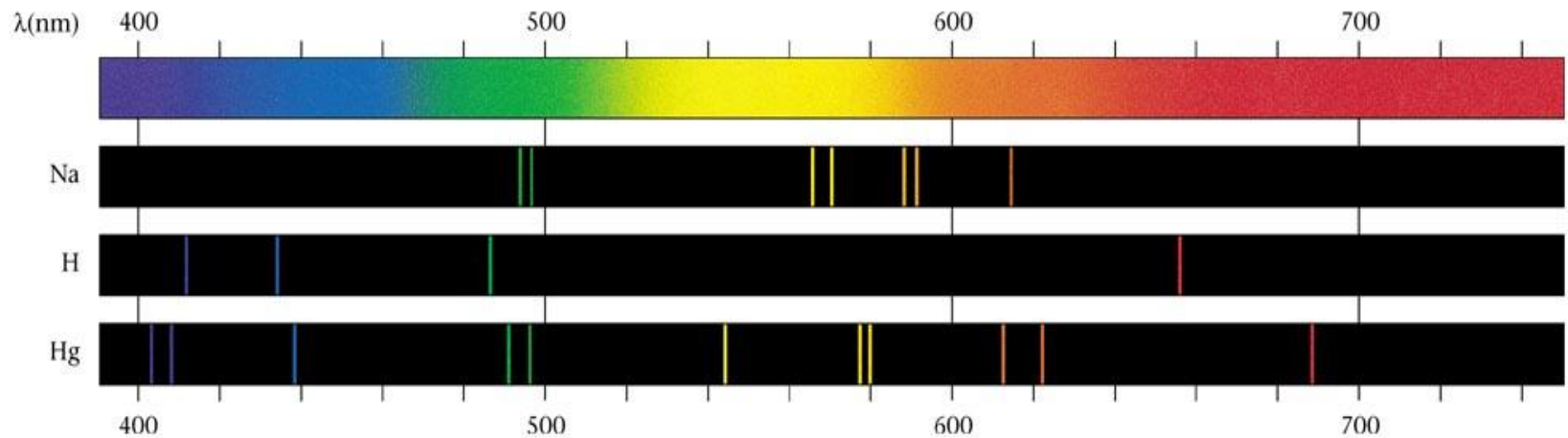
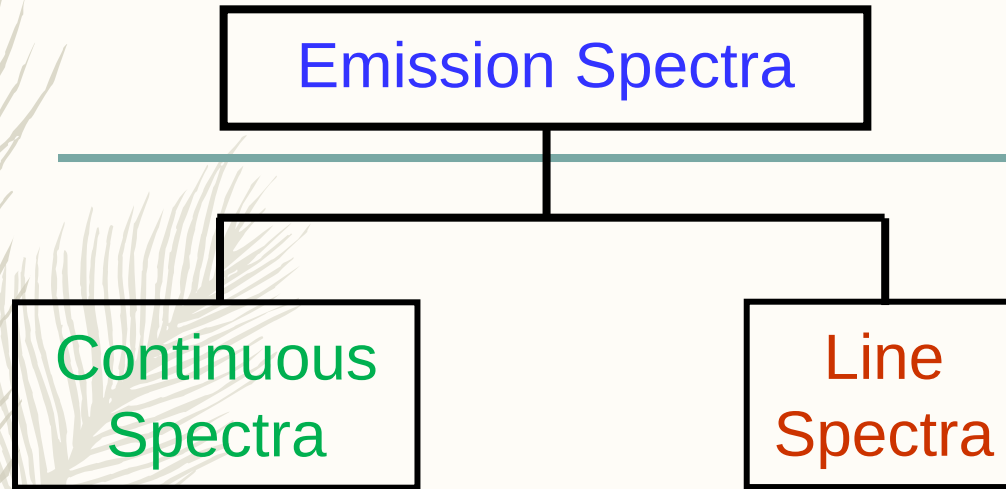
$$E_{\text{photon}} = h \times c / \lambda$$

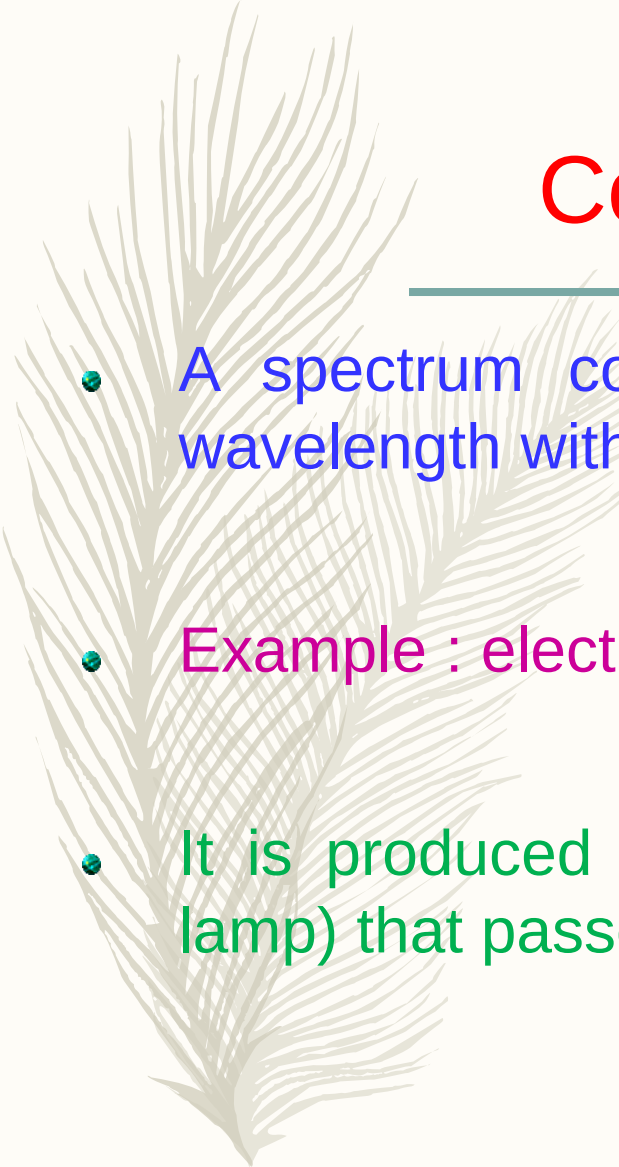
$$\lambda = h \times c / E_{\text{photon}}$$

$$\lambda = 6.63 \times 10^{-34} \text{ (J}\cdot\text{s)} \times 3.00 \times 10^8 \text{ (m/s)} / 1.55 \times 10^{-19} \text{ J}$$

$$\lambda = 1280 \text{ nm}$$

Emission Spectra

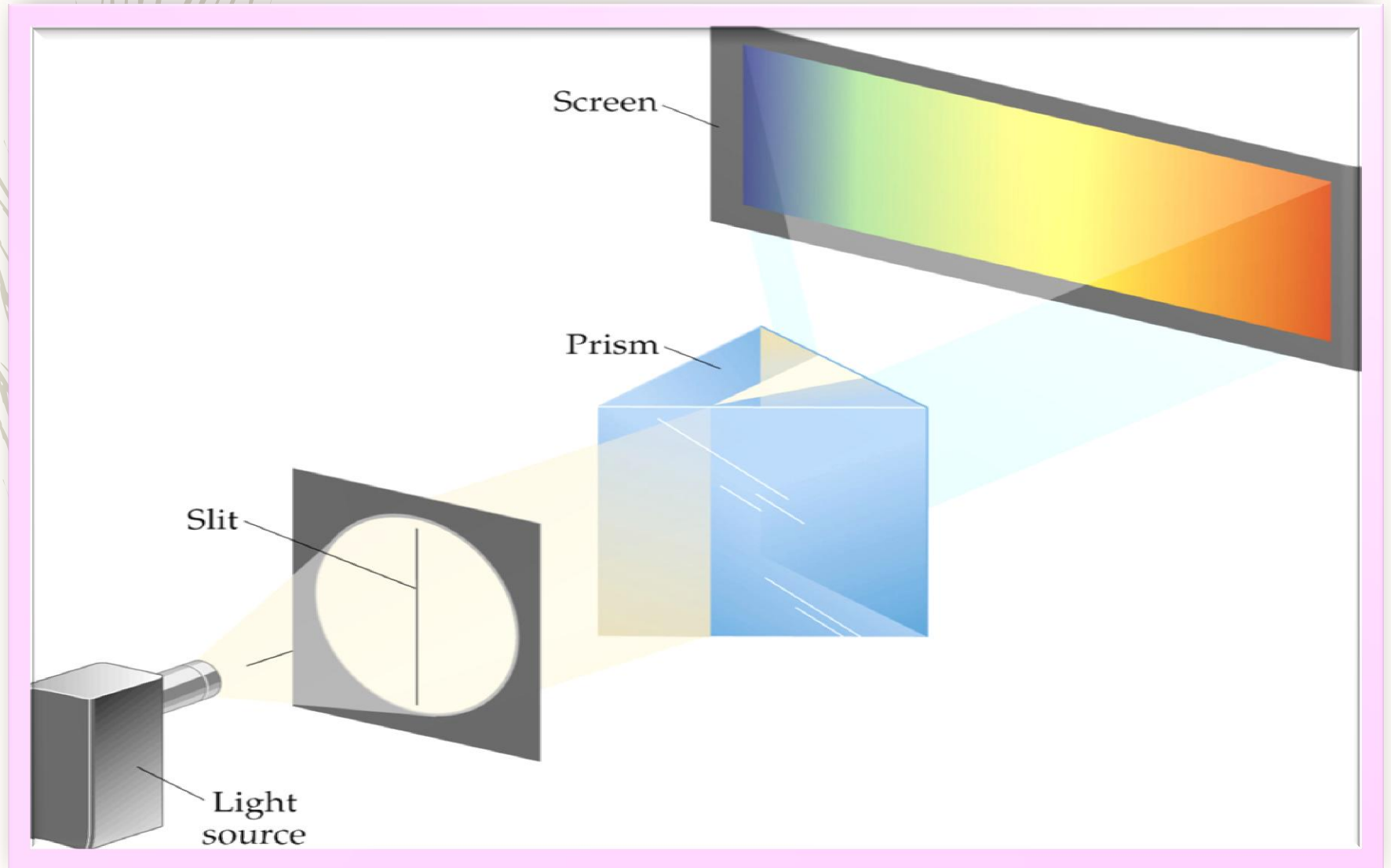




Continuous Spectrum

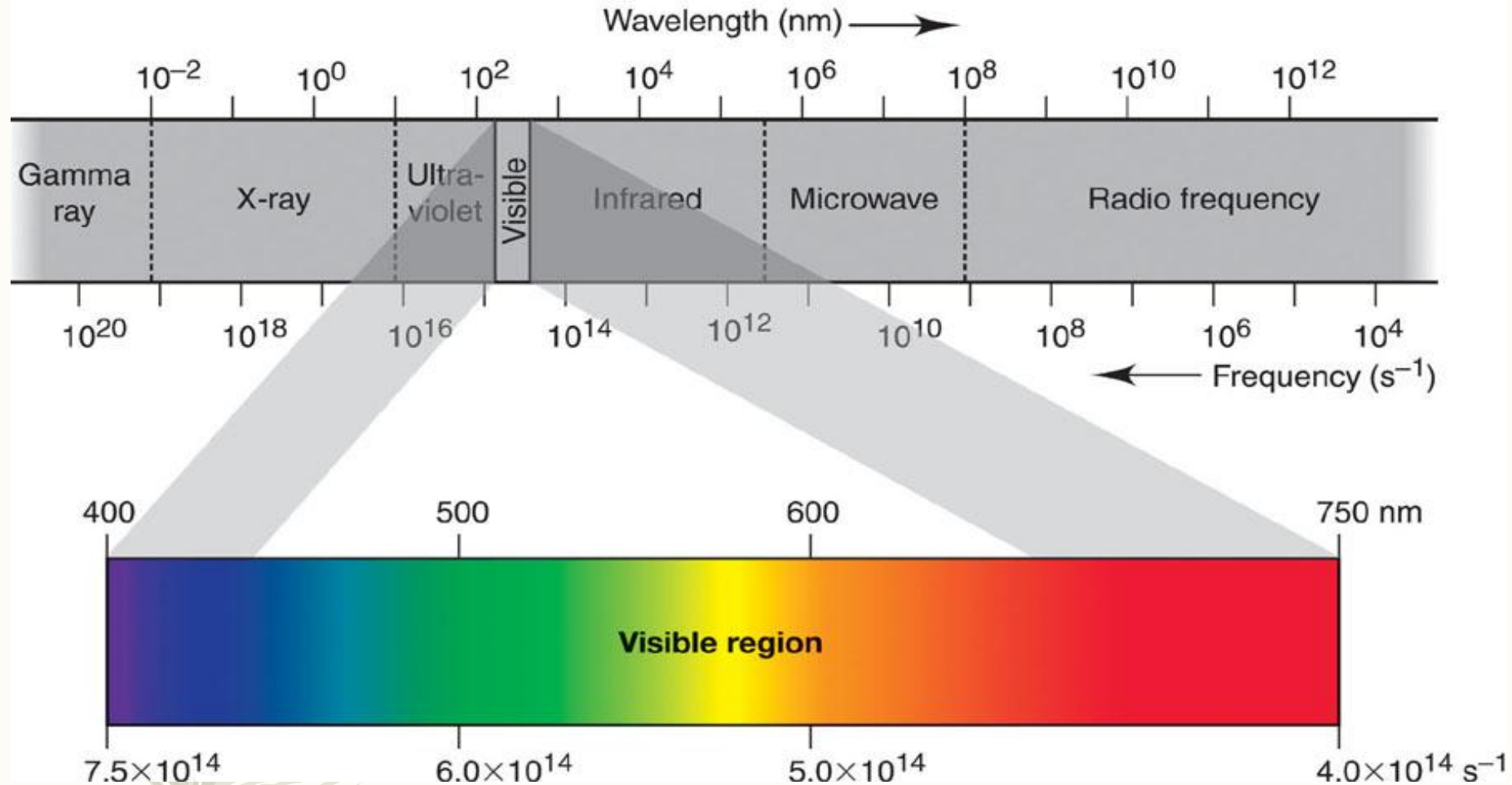
- A spectrum consists of radiation distributed over all wavelength without any blank spot.
- Example : electromagnetic spectrum, rainbow
- It is produced by white light (sunlight or incandescent lamp) that passed through a prism

Formation of Continuous Spectrum



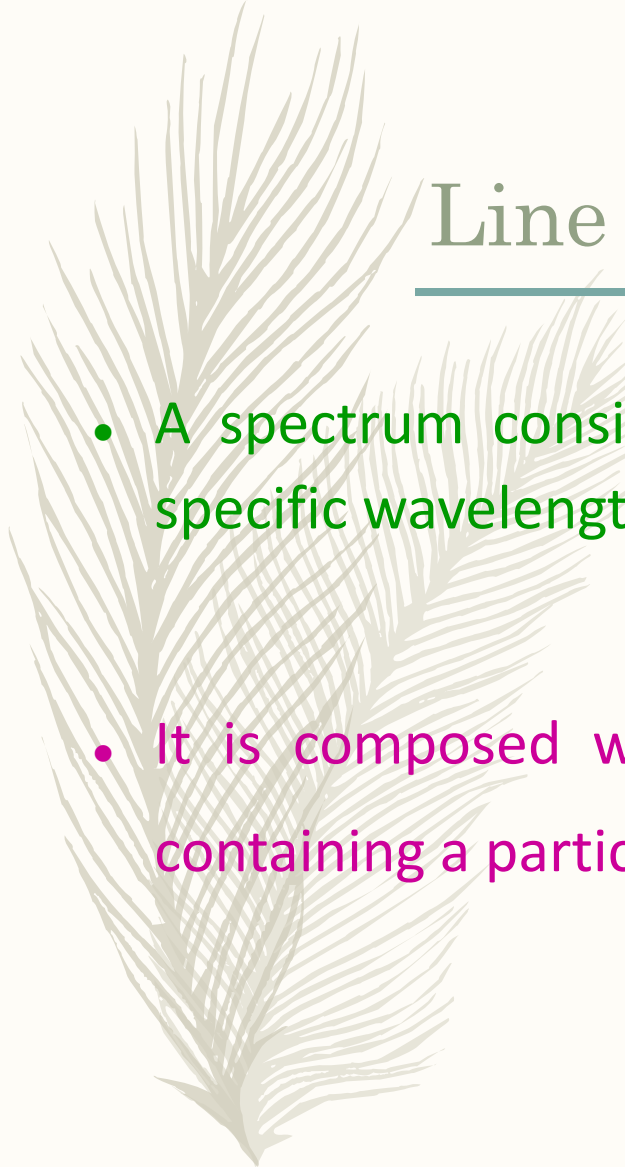
Regions of the Electromagnetic Spectrum

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



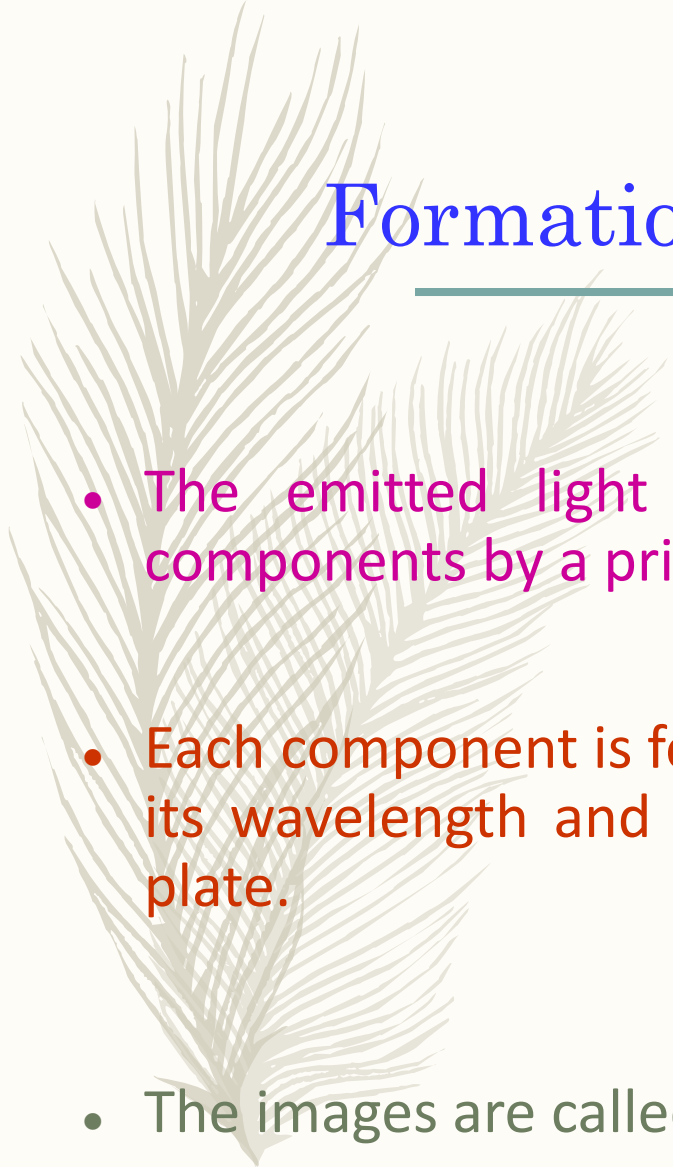
Shortest wavelength,
highest frequency

Longest wavelength,
lowest frequency



Line Spectrum (Atomic Spectrum)

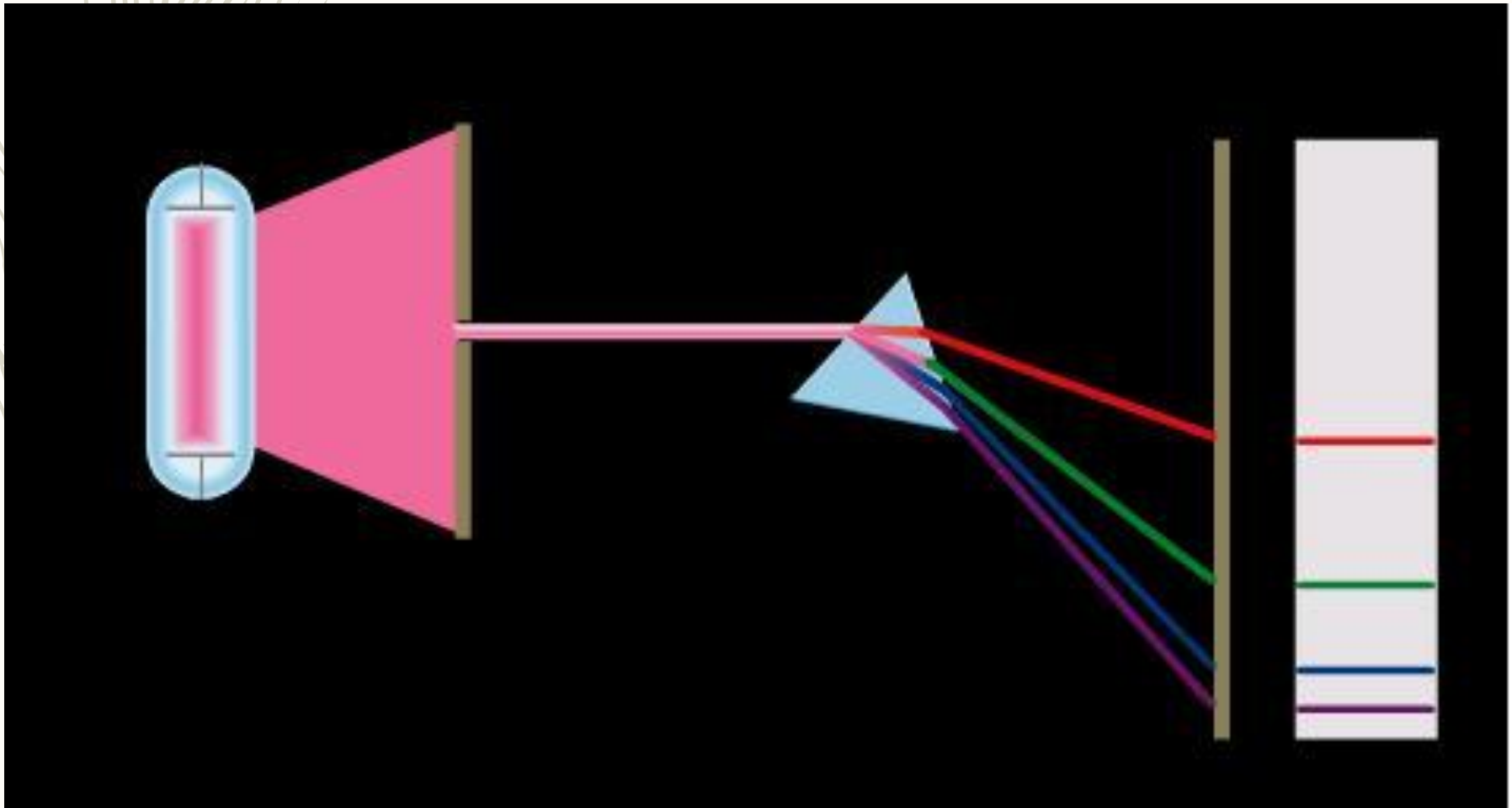
- A spectrum consists of discontinuous & discrete lines with specific wavelength.
- It is composed when the light from a gas discharge tube containing a particular element is passed through a prism.



Formation of Atomic / Line Spectrum

- The emitted light (photons) is then separated into its components by a prism.
- Each component is focused at a definite position, according to its wavelength and forms as an image on the photographic plate.
- The images are called spectral lines.

Formation of Atomic / Line Spectrum



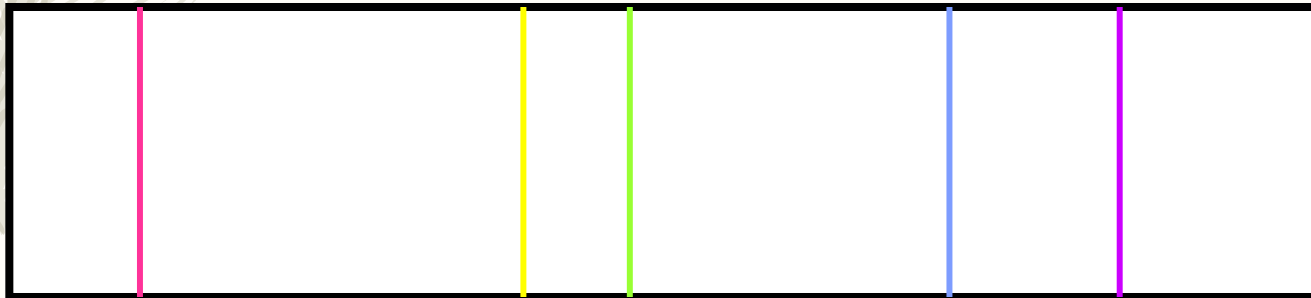
Formation of Atomic / Line Spectrum

- Example : The line emission spectrum of hydrogen atom.

- Line spectrum are composed a few wavelengths giving a series of discrete line separated by blank areas.
- Each line corresponds to a specific wavelength or frequency.

Formation of Line Spectrum

- When electron absorbed radiant energy, they will move from lower energy level to higher energy level (excited state).
- This excited electron is unstable and it will fall back to lower energy level.
- During the transition, electron will release energy in the form of light with specific wavelength and can detected as a line spectrum.



Differences Between Line & Continuous Spectra

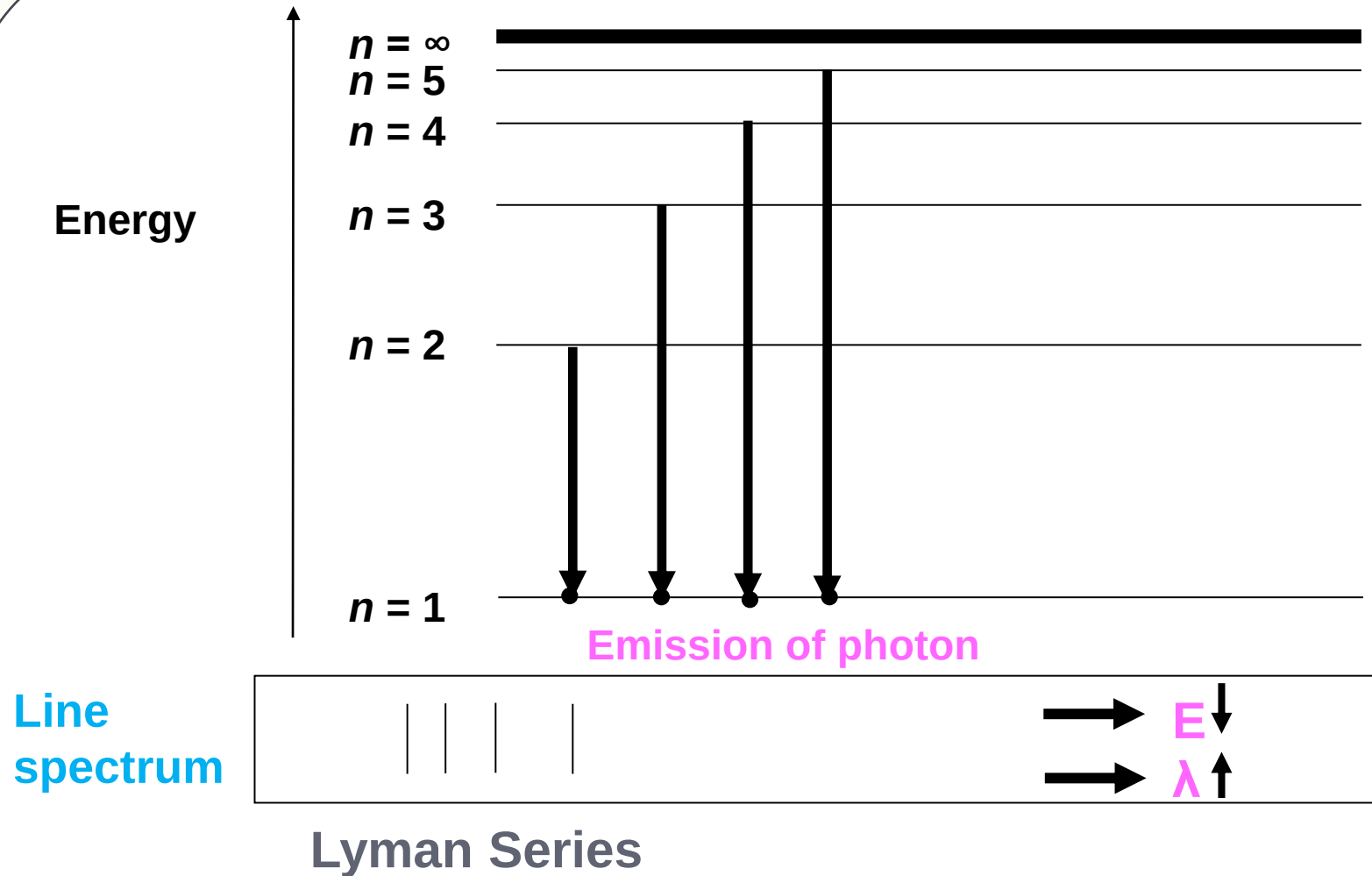
Continuous Spectrum

- A spectrum that contains all wavelength without any blank spots.
- Example: Rainbow.

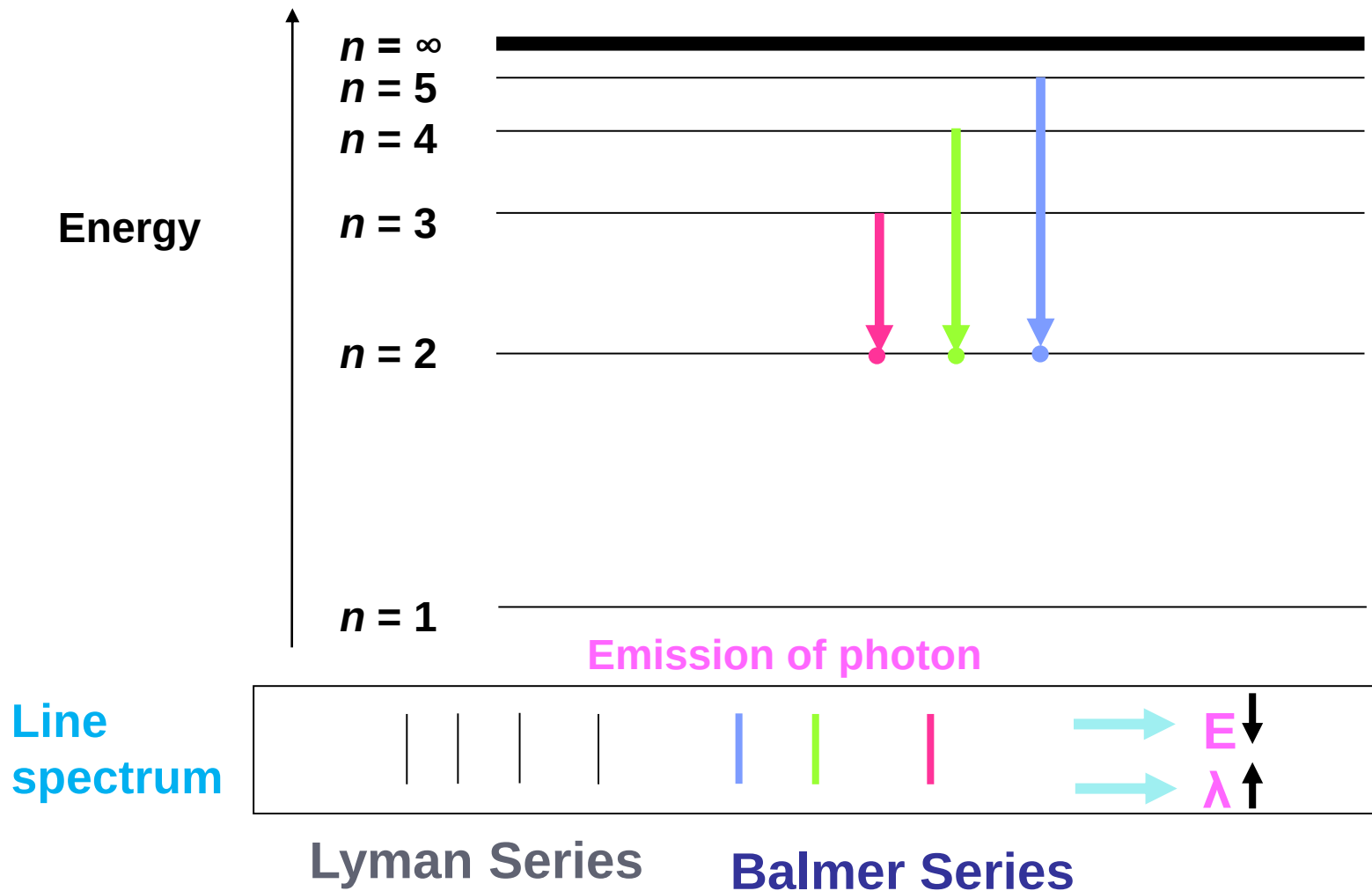
Line Spectrum

- A spectrum that contain only specific wavelengths.
- A spectrum of discrete lines with certain wavelengths.
- Example: Emission spectrum an element.

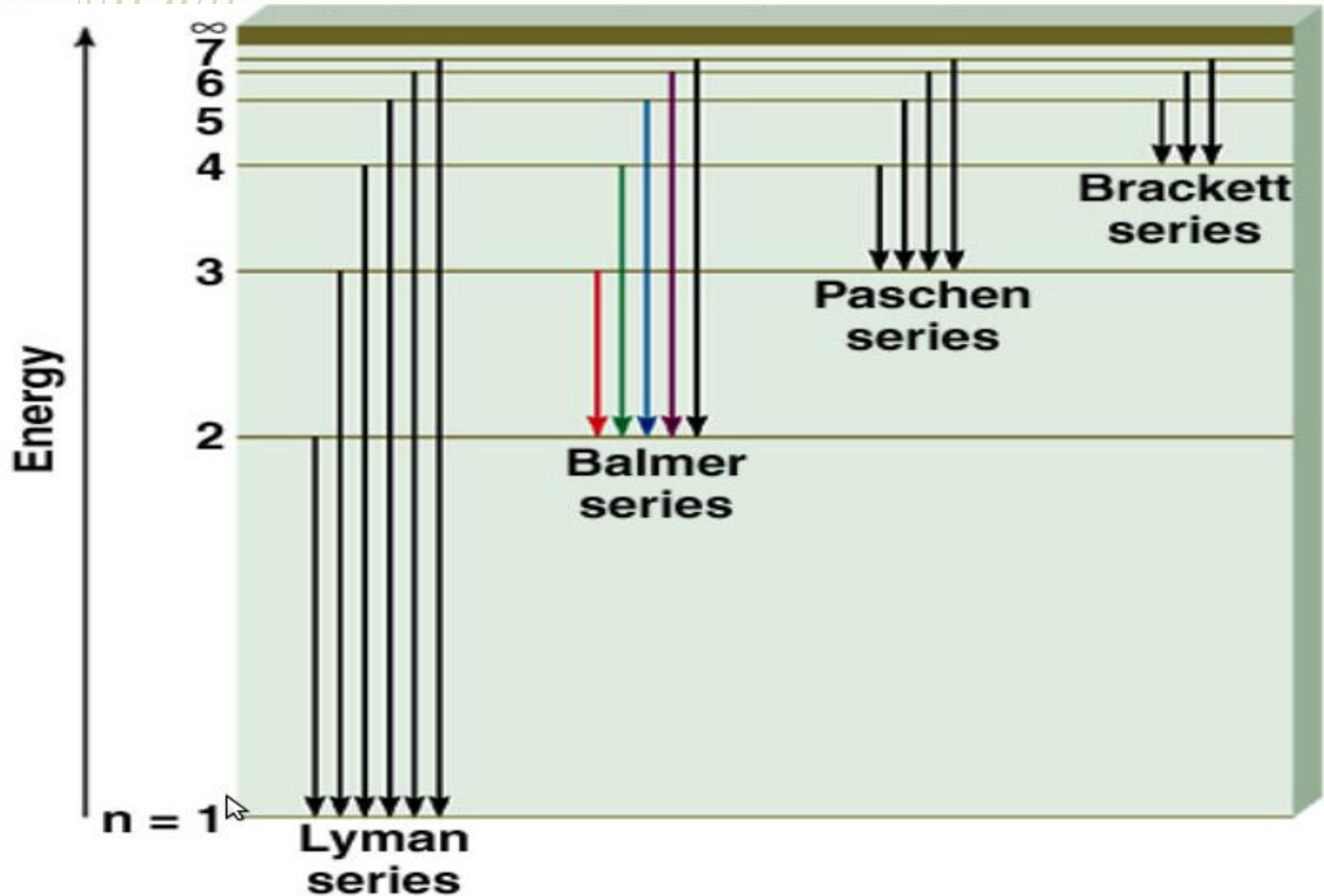
Formation of Line Spectrum (Lyman Series)



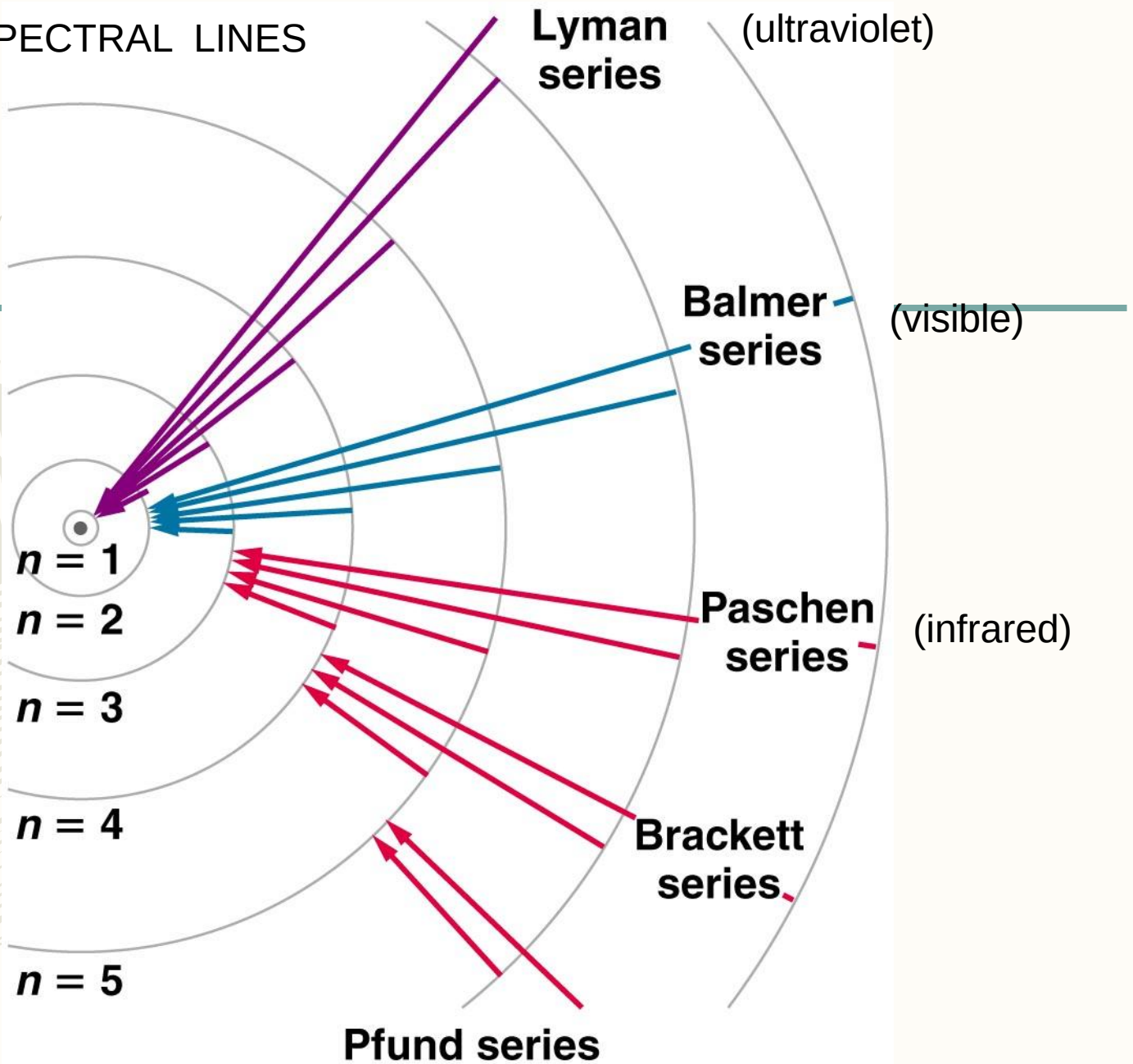
Formation of Line Spectrum (Balmer Series)



Energy Level in Hydrogen Atom



HYDROGEN SPECTRAL LINES

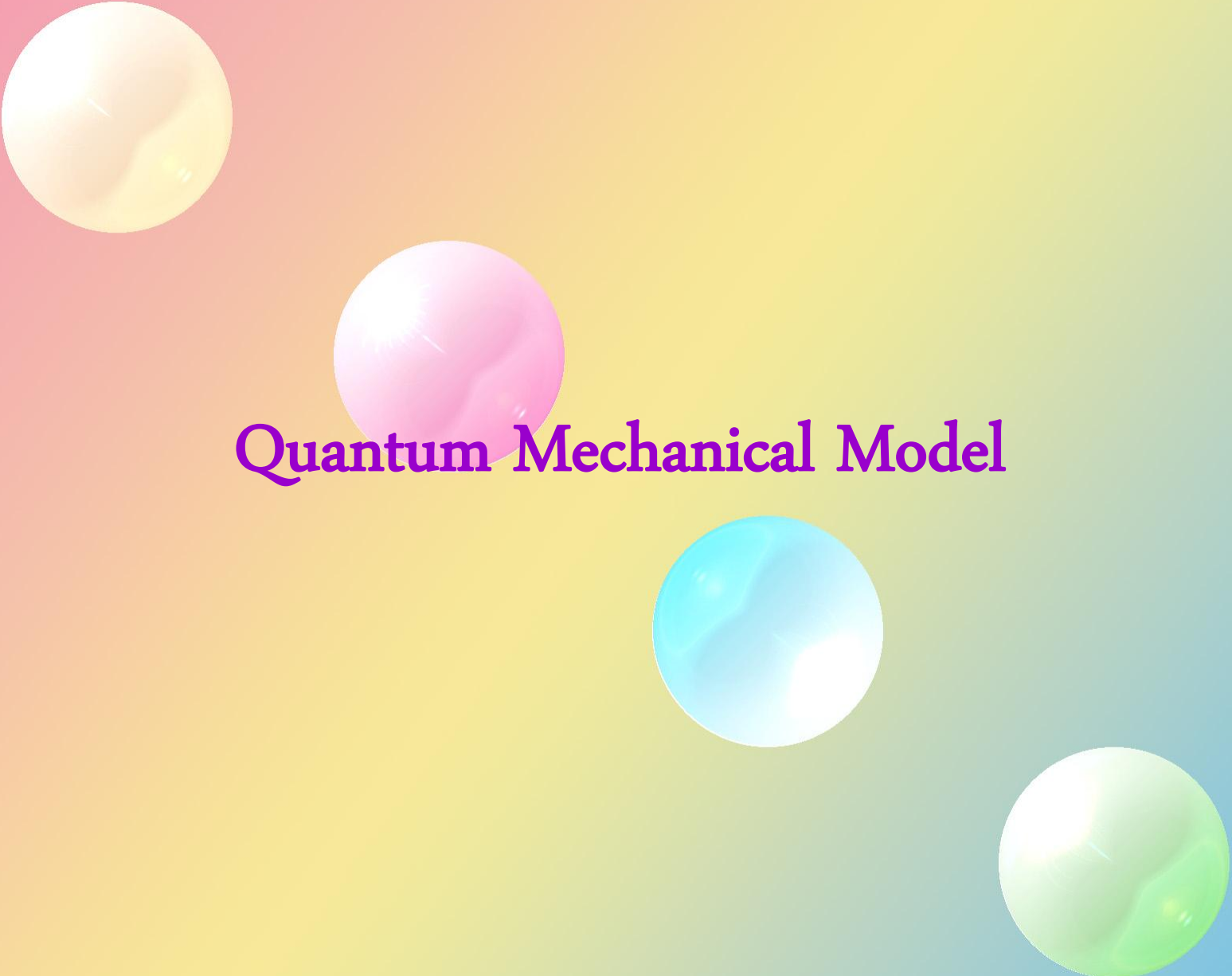


1) What is the wavelength of a photon (in nm) emitted during transition from $n_i = 5$ state to the $n_f = 2$ state in hydrogen atom? (434 nm)

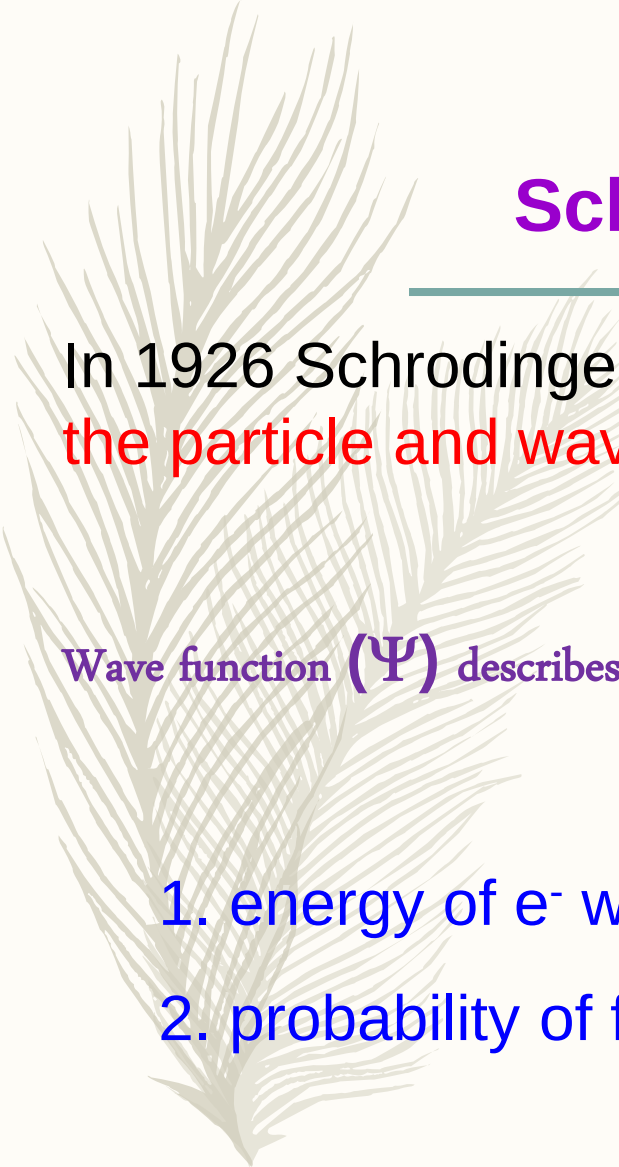
2) What is the wavelength (in nm) of a photon emitted during a transition from $n_i = 6$ state to the $n_f = 4$ state in H atom? (2626 nm)

3) Which transition in the hydrogen atom would emit light of a shorter wavelength?

- i) $n_i = 5 \rightarrow n_f = 3$
- ii) $n_i = 4 \rightarrow n_f = 2$



Quantum Mechanical Model



Schrodinger Wave Equation

In 1926 Schrodinger wrote an equation that described both the particle and wave nature of the e^-

Wave function (Ψ) describes:

1. energy of e^- with a given Y
2. probability of finding e^- in a volume of space

According to Schrodinger,

- electrons are treated as a matter as well as waves.

- electrons are located in an orbitals.

- ➡Orbitals : region where there is a high probability of finding an electron.

- The concept of orbitals is, that not all electrons in an atom are located at the same distance from the nucleus.

- They located at certain specified space from the nucleus where the probability of finding electrons is the greatest.

- Based on Schrodinger, orbitals are described in mathematical terms as having wave functions that is characterized by the values of four quantum numbers.

Quantum Numbers

Each of the electrons in an atom is described and characterized by set of four quantum numbers, namely

- principal quantum number, n
- angular momentum quantum number, l
- magnetic quantum number, m
- electron spin quantum number, s .

Quantum numbers

Schrodinger Wave Equation

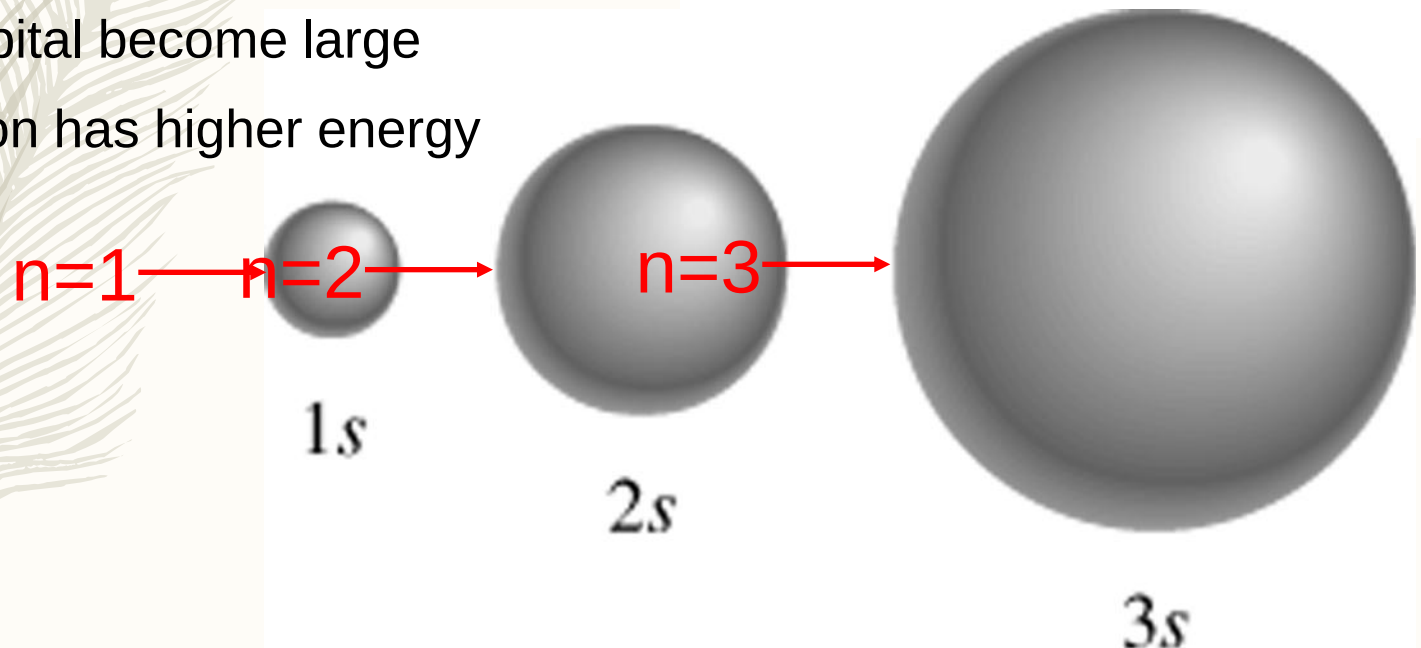
$$\Psi = \text{fn}(\mathbf{n}, l, m_l, m_s) \quad \text{Principal Quantum Number, } n$$

n determines the energy level (electron shell) and size of an orbital

$$n = 1, 2, 3, 4, \dots$$

As n increase :

- the orbital become large
- electron has higher energy



Schrodinger Wave Equation

$$\Psi = \text{fn}(n, l, m_l, m_s)$$

Angular momentum quantum number, l

Indicates the shape of the atomic orbitals

For a given value of n , $l = 0, 1, 2, 3, \dots n-1$

$n = 1, l = 0$
 $n = 2, l = 0 \text{ or } 1$
 $n = 3, l = 0, 1, \text{ or } 2$

$l = 0$	s orbital
$l = 1$	p orbital
$l = 2$	d orbital
$l = 3$	f orbital

Each letter
correspond
to different
sublevel and
differently
shaped
orbital

Angular Momentum Quantum Number, l

- Value is depend on n . (i.e., $0 \leq l < n$).

If $n = 1$, $l = 0$ (s-orbital)

One subshell (s orbital)

If $n = 2$, $l = 0$ (s-orbital)

$l = 1$ (p-orbital)

Two subshells (s and p orbitals)

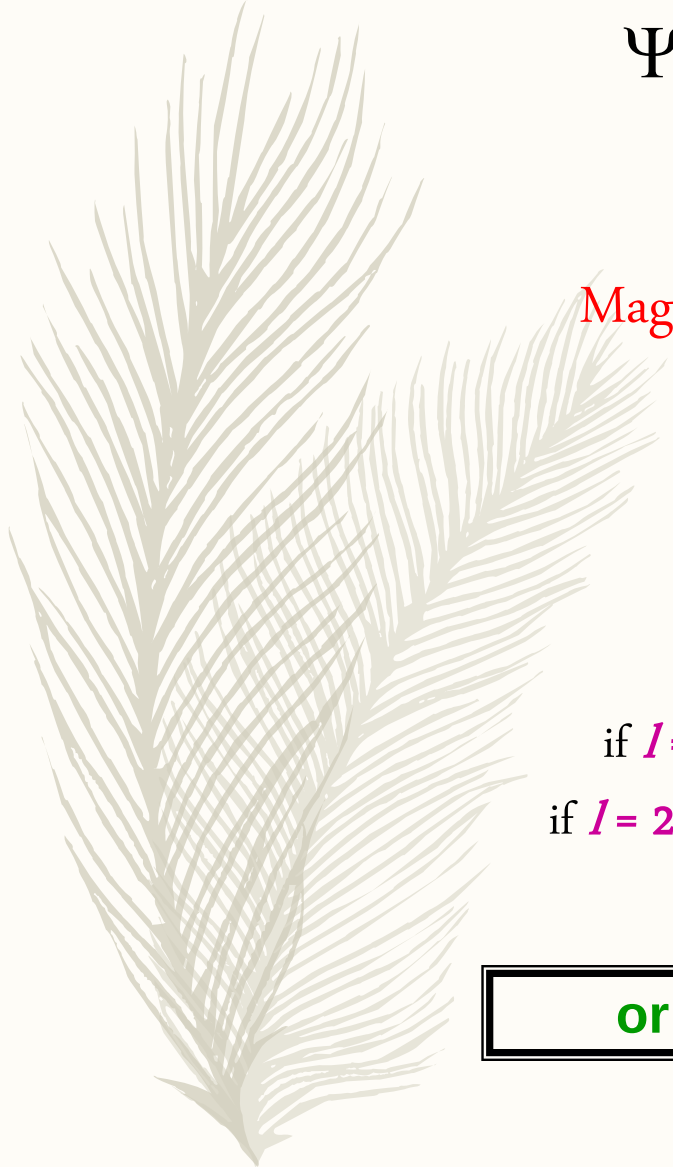
If $n = 3$, $l = 0$ (s-orbital)

$l = 1$ (p-orbital)

$l = 2$ (d-orbital)

Three subshells (s, p, and d orbitals)

Within a shell different
sublevel / subshell are
possible where each has a
characteristic shape.


$$\Psi = \text{fn}(n, l, m_l, m_s)$$

Magnetic quantum number, m_l

for a given value of l

$$m_l = -l, \dots, 0, \dots, +l$$

if $l = 1$ (p orbital), $m_l = -1, 0, \text{ or } 1$

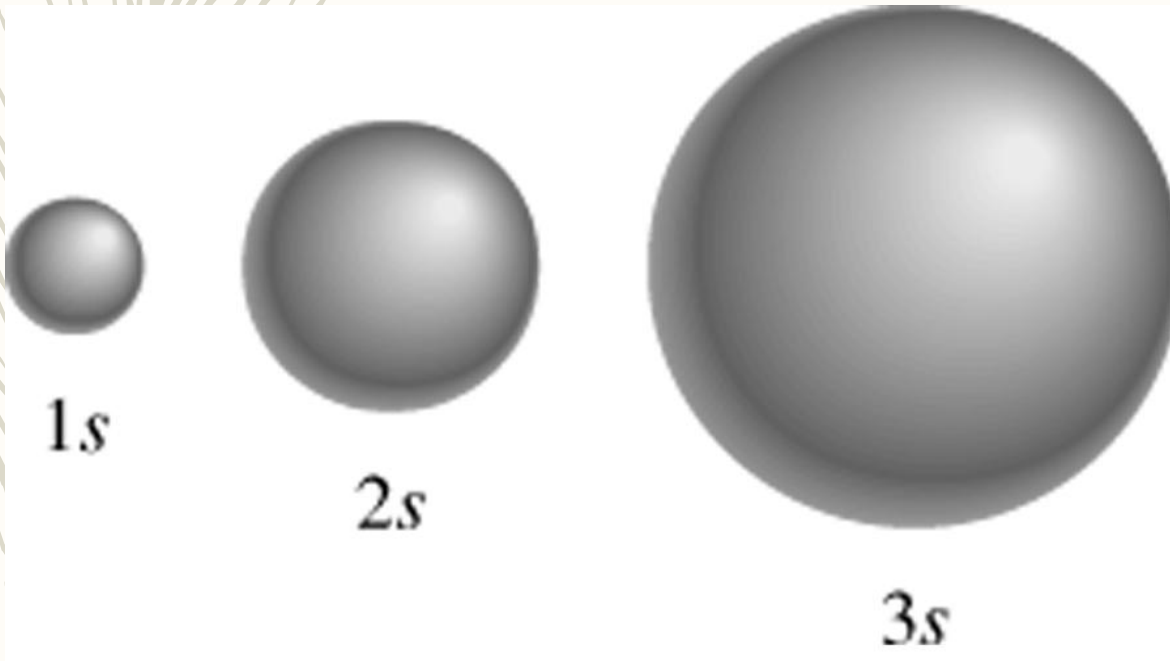
if $l = 2$ (d orbital), $m_l = -2, -1, 0, 1, \text{ or } 2$

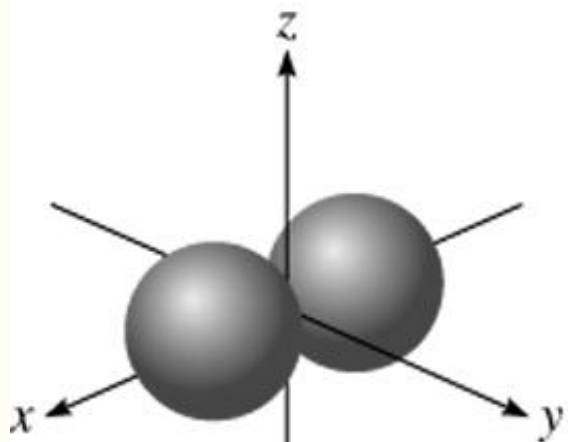
m_l designates a
specific orbital
within a
subshell

orientation of the orbital in space

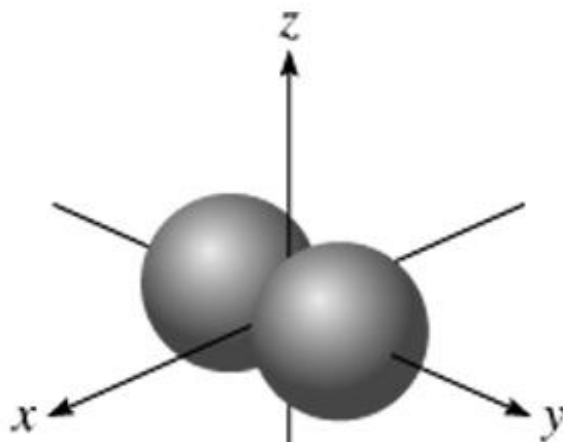
orbitals-shapes

$l = 0$ (s orbitals)

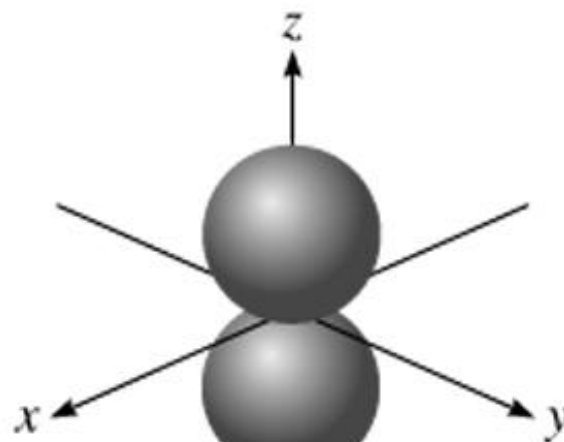




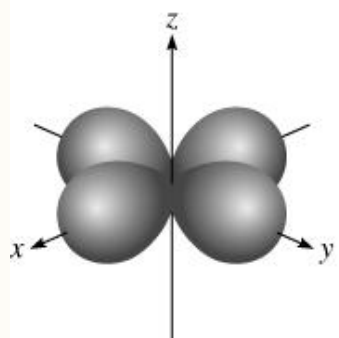
$$m_l = -1$$

 $2p_x$


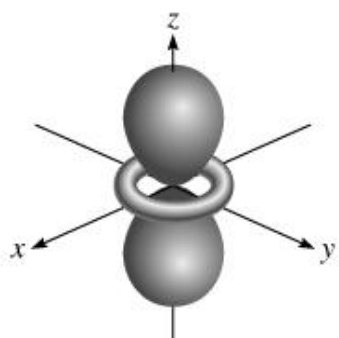
$$m_l = 0$$

 $2p_y$


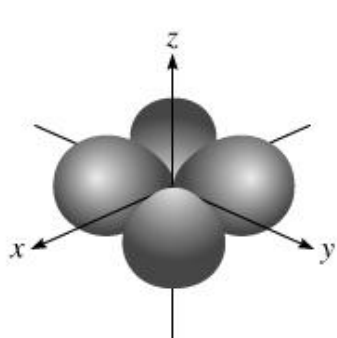
$$m_l = 1$$

 $2p_z$

 $3d_{x^2-y^2}$

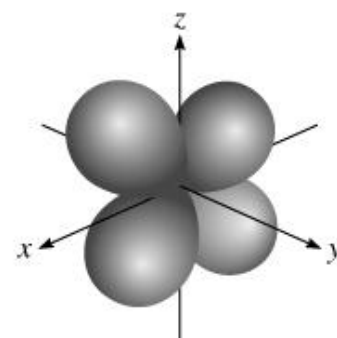
$$m_l = -2$$


 $3d_{z^2}$

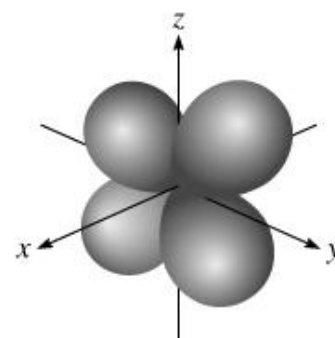
$$m_l = -1$$


 $3d_{xy}$

$$m_l = 0$$


 $3d_{xz}$

$$m_l = 1$$


 $3d_{yz}$

$$m_l = 2$$

TABLE 6.2 Relationship among Values of n , l , and m_l through $n = 4$

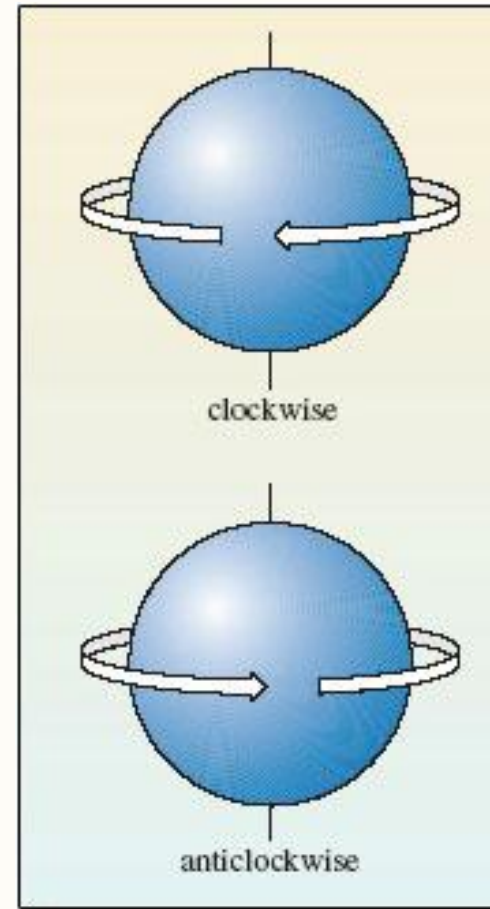
n	Possible Values of l	Subshell Designation	Possible Values of m_l	Number of Orbitals in Subshell	Total Number of Orbitals in Shell
1	0	1s	0	1	1
2	0	2s	0	1	4
	1	2p	1, 0, -1	3	
3	0	3s	0	1	9
	1	3p	1, 0, -1	3	
	2	3d	2, 1, 0, -1, -2	5	
4	0	4s	0	1	16
	1	4p	1, 0, -1	3	
	2	4d	2, 1, 0, -1, -2	5	
	3	4f	3, 2, 1, 0, -1, -2, -3	7	

Schrodinger Wave Equation

$$\Psi = \text{fn}(n, l, m_l, m_s)$$

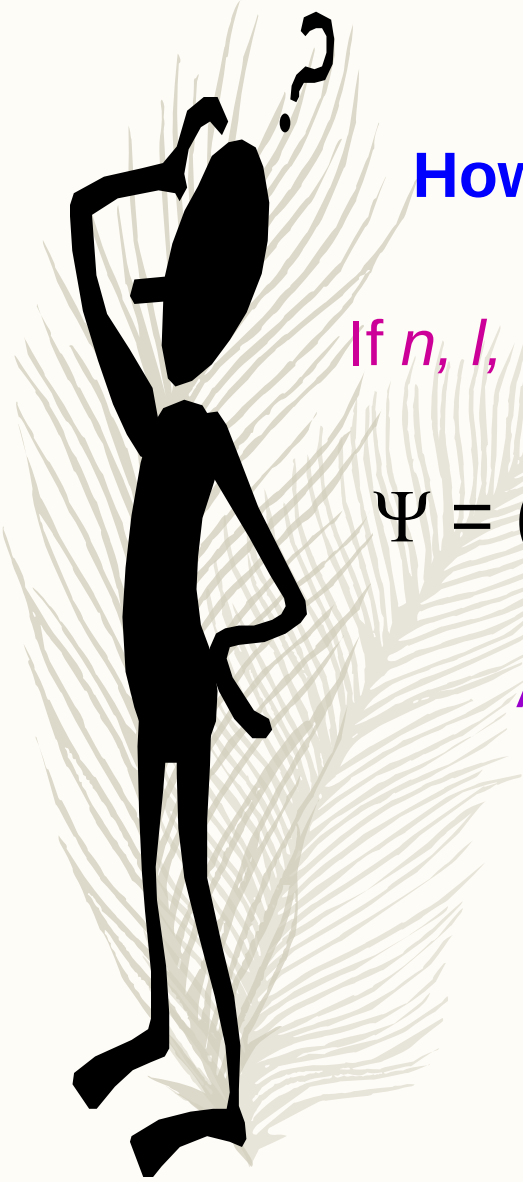
spin quantum number m_s

- The value of s represent the direction of an electron rotation on its own axis.
- either clockwise or anticlockwise
- It has 2 value : $+1/2$ and $-1/2$



$$m_s = +1/2$$

$$m_s = -1/2$$



How many electrons can an orbital hold?

If n , l , and m_l are fixed, then $m_s = \frac{1}{2}$ or $-\frac{1}{2}$

$$\Psi = (n, l, m_l, \frac{1}{2}) \text{ or } \Psi = (n, l, m_l, -\frac{1}{2})$$

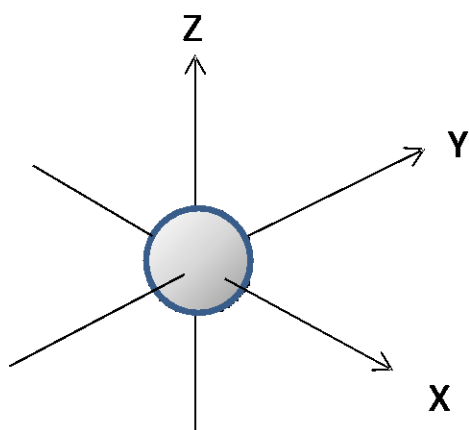
An orbital can hold 2 electrons

Shape of Atomic Orbital

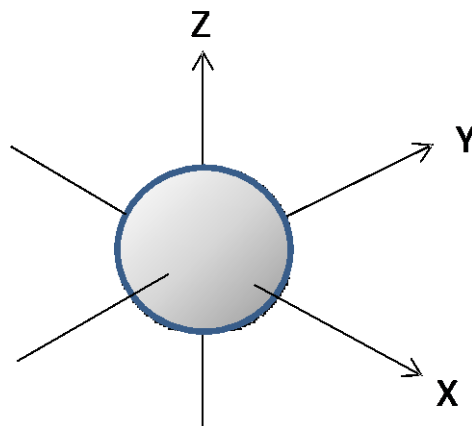
s orbitals

- Spherical shape with the nucleus at the center.
- When $l = 0$, $m = 0$, only 1 orientation of s orbital.
- The larger value of n , the size of s orbital gets larger.

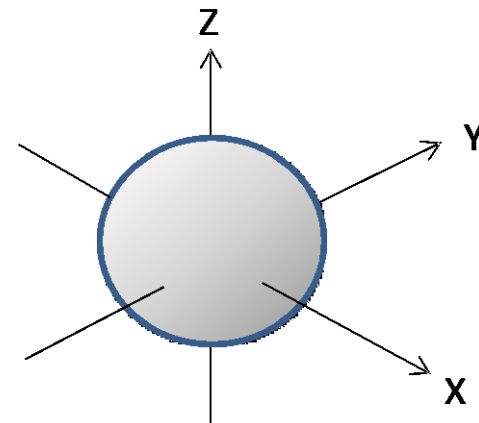
Shape of s orbital with different n



1s



2s



3s

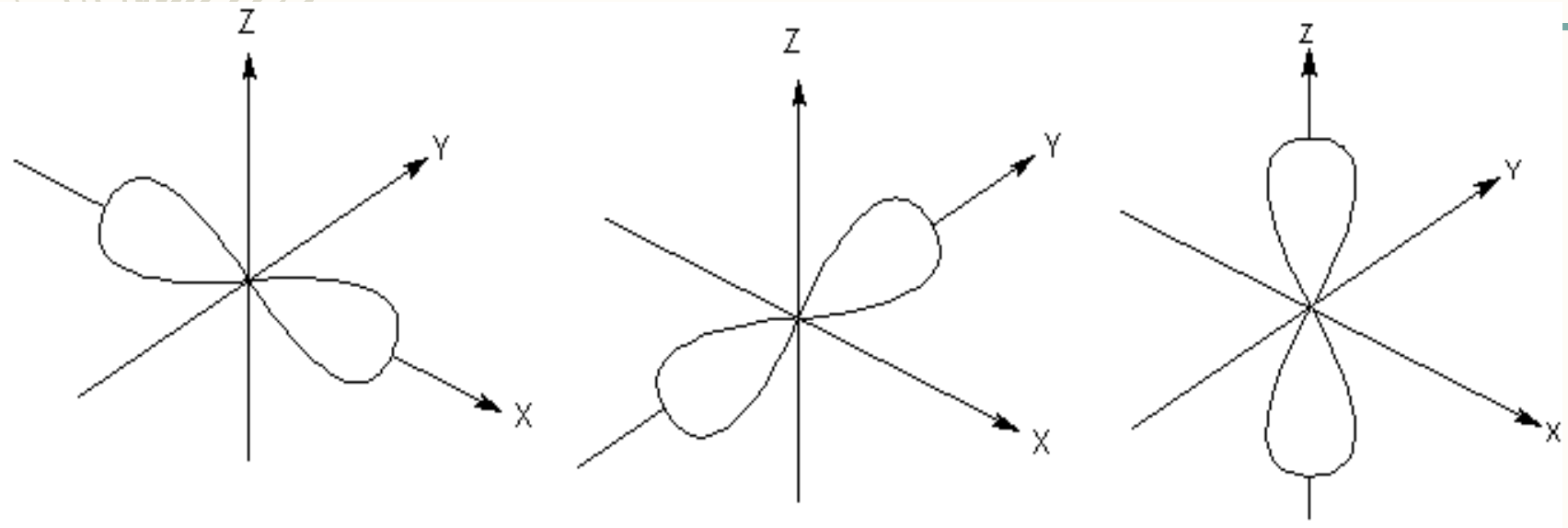


Shape of Atomic Orbitals

p orbitals

- Can be represent as a pair of dumb-bell shaped
- When $l = 1$, $m = -1, 0, +1$
- 3 orientation of p-orbitals p_x , p_y , and p_z .
- As n increases, the p orbitals get larger.

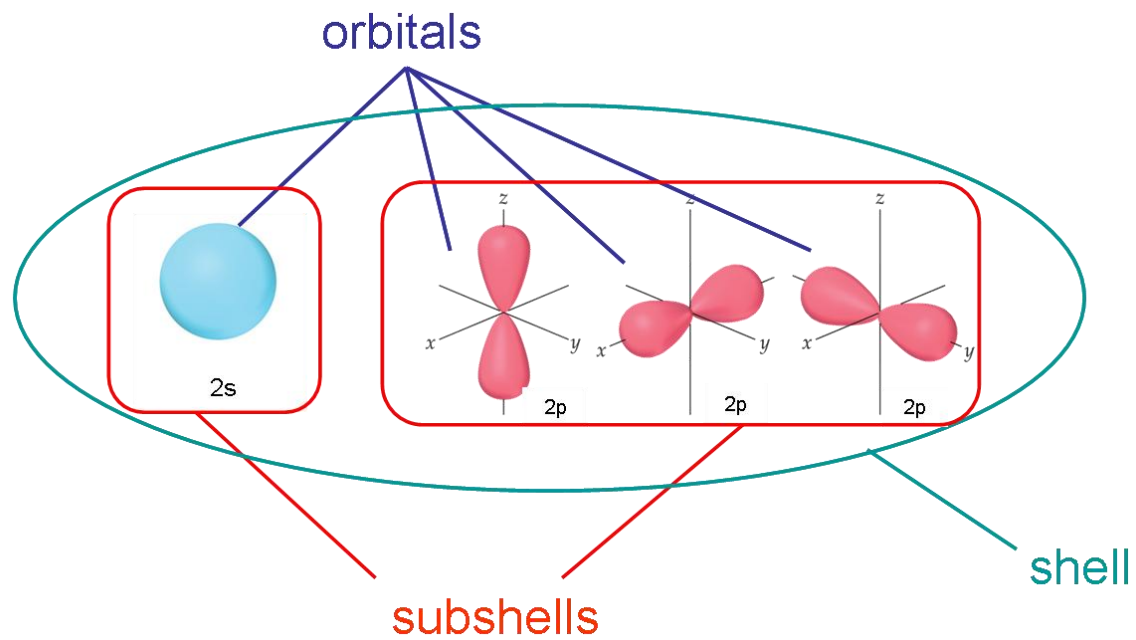
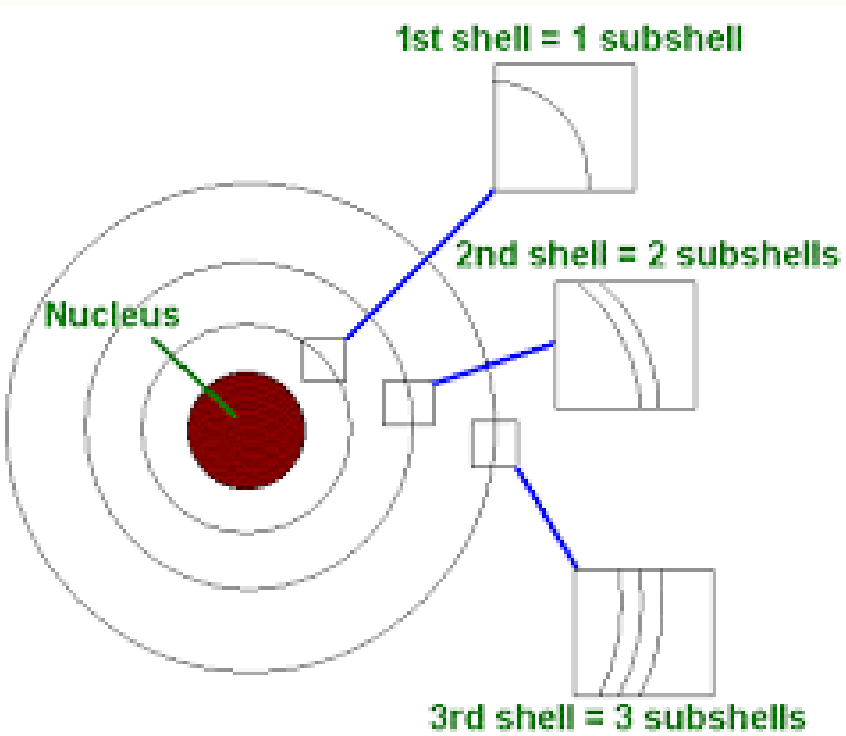
Shape of p orbital



p_x

p_y

p_z

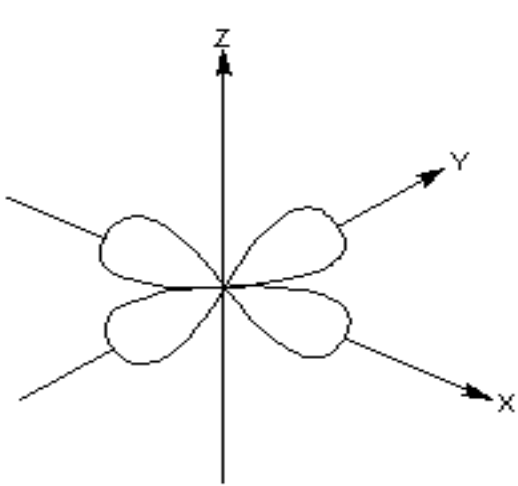


Shape of Atomic Orbitals

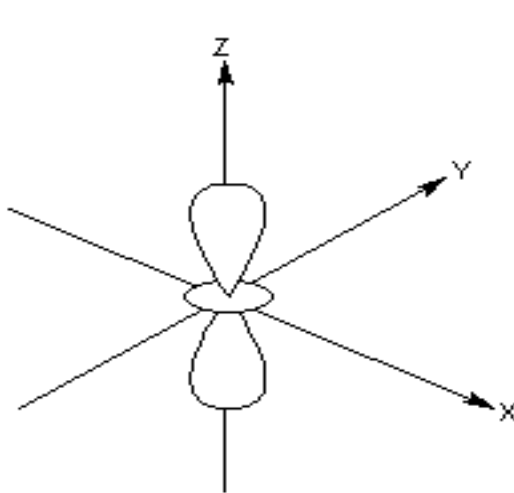
d orbitals

- All the d orbitals do not look alike.
- When $l = 2$, $m = -2, -1, 0, +1, +2$.
- There are five orientation of d orbitals.

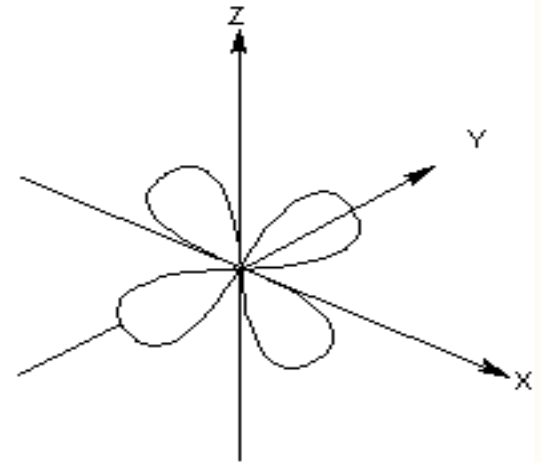
Shape of d orbital



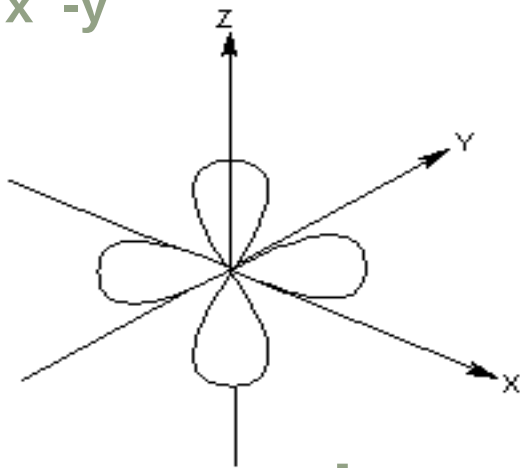
$d_{x^2-y^2}$



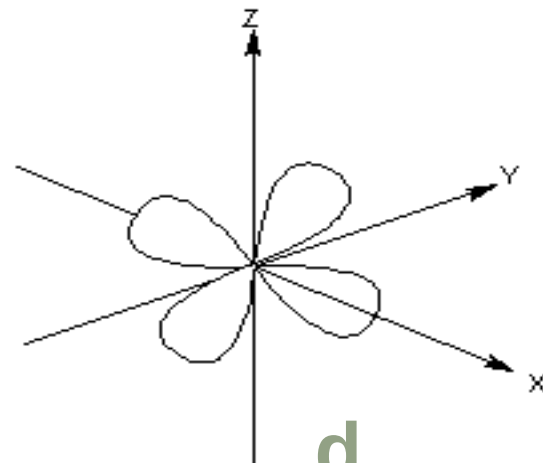
d_{z^2}



d_{xy}



d_{xz}



d_{yz}



How many 2p orbitals are there in an atom?

$n=2$



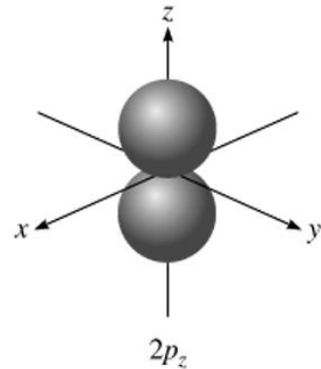
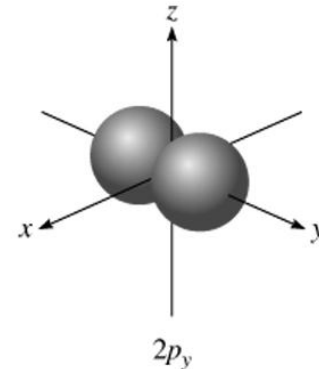
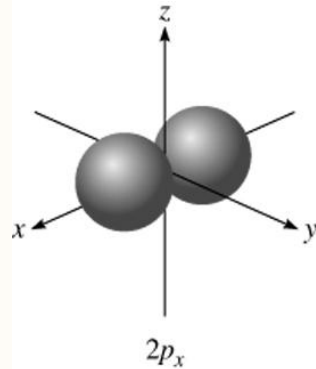
2p



$l=1$

If $l = 1$, then $m_l = -1, 0, \text{ or } +1$

3 orbitals



How many electrons can be placed in the 3d subshell?

$n=3$



3d



$l=2$

If $l = 2$, then $m_l = -2, -1, 0, +1, \text{ or } +2$

5 orbitals which can hold a total of 10 e^-



Try this...

1. List the values of n , l and m_l for orbitals in the 4d subshell.
2. What is the total number orbitals associated with the principal quantum number $n = 3$?

3p and 3d Orbitals

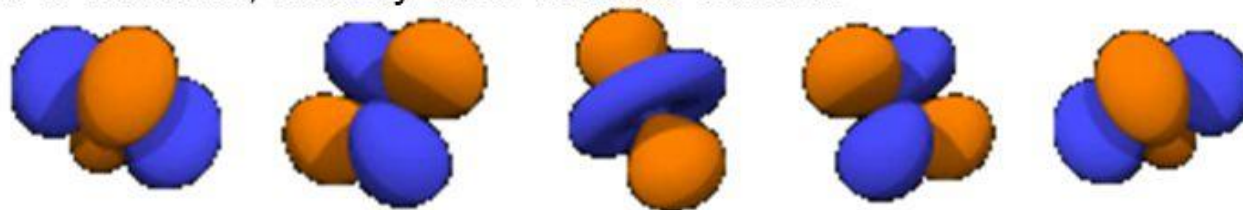
When $n = 3$, then l can be 0 (one 3s orbital),

$l = 1$ (three 3p orbitals)

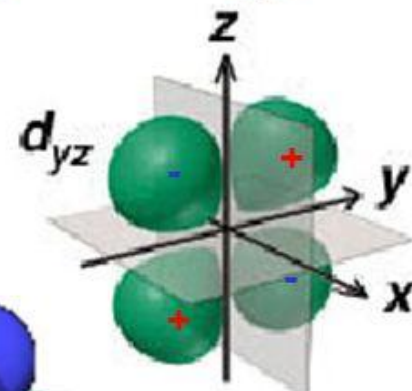
3p orbitals have the same shape and designation as 2p orbitals ($3p_x$, $3p_y$, $3p_z$), but have an extra spherical node.

or $l = 2$ (d-orbitals). 3d orbitals have more lobes than 2p orbitals, and their shape is obviously different.

For d-orbitals, m may take on five values:



Higher quantum number n leads to more nodes in all orbitals.



3d orbitals have four lobes. The lobes point along pair of axes ($d_{x^2-y^2}$), or between axes (d_{xy} , d_{xz} , d_{yz}) or along the z axis (d_{z^2}).

Questions

a. What is the maximum number of electrons that can be present in the principal level for which $n = 3$?

b. An oxygen atom has a total of eight electrons. Write the four quantum numbers for each of the eight electrons in the ground state.



An oxygen atom has a total of eight electrons. Write the four quantum numbers for each of the eight electrons in the ground state.

- Assuming you know the basics behind [quantum numbers](#), you can write the unique 4-number set for each electron belonging to oxygen like this
- *The first energy level:*
 - $n=1, l=0, m_l=0, m_s=-1/2 \rightarrow$ the electron located in the **1s-orbital**, *spin-down*;
 - $n=1, l=0, m_l=0, m_s=+1/2 \rightarrow$ the electron located in the **1s-orbital**, *spin-up*;
- *The second energy level:*
 - $n=2, l=0, m_l=0, m_s=-1/2 \rightarrow$ the electron located in the **2s-orbital**, *spin-down*;
 - $n=2, l=0, m_l=0, m_s=+1/2 \rightarrow$ the electron located in the **2s-orbital**, *spin-up*;
 - $n=2, l=1, m_l=-1, m_s=-1/2 \rightarrow$ the electron located in the 2p_x-orbital, *spin-down*;
 - $n=2, l=1, m_l=-1, m_s=+1/2 \rightarrow$ the electron located in the 2p_x-orbital, *spin-up*;
 - $n=2, l=1, m_l=0, m_s=+1/2 \rightarrow$ the electron located in the 2p_y-orbital, *spin-up*;
 - $n=2, l=1, m_l=1, m_s=+1/2 \rightarrow$ the electron located in the 2p_z-orbital, *spin-up*;

Answer:

Quantum numbers are used to describe the position of an electron in an atom.

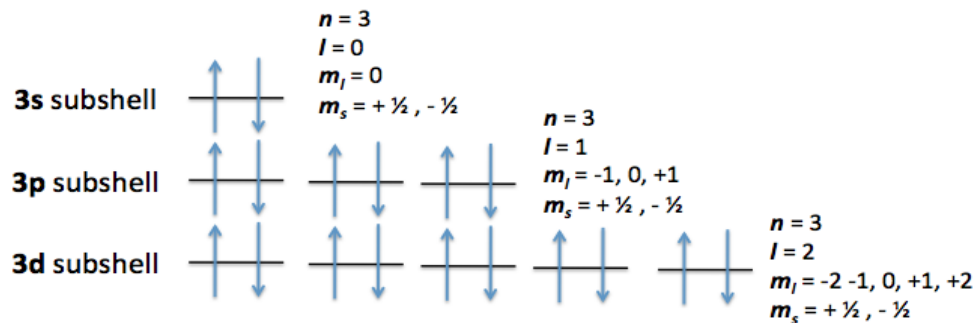
Explanation: Quantum numbers can be used to describe the exact location of an electron.

There are four quantum numbers:

1. n - describes the energy level
2. ℓ - describes the subenergy level (**0 to $n-1$**)
3. m_ℓ - describes the orbital (**$-\ell$ to $+\ell$**)
4. m_s - describes the spin (clockwise or anti clockwise : $-\frac{1}{2}$ or $+\frac{1}{2}$)

Example:

Electron Orbital Diagram of subshell/orbital combinations when $n = 3$

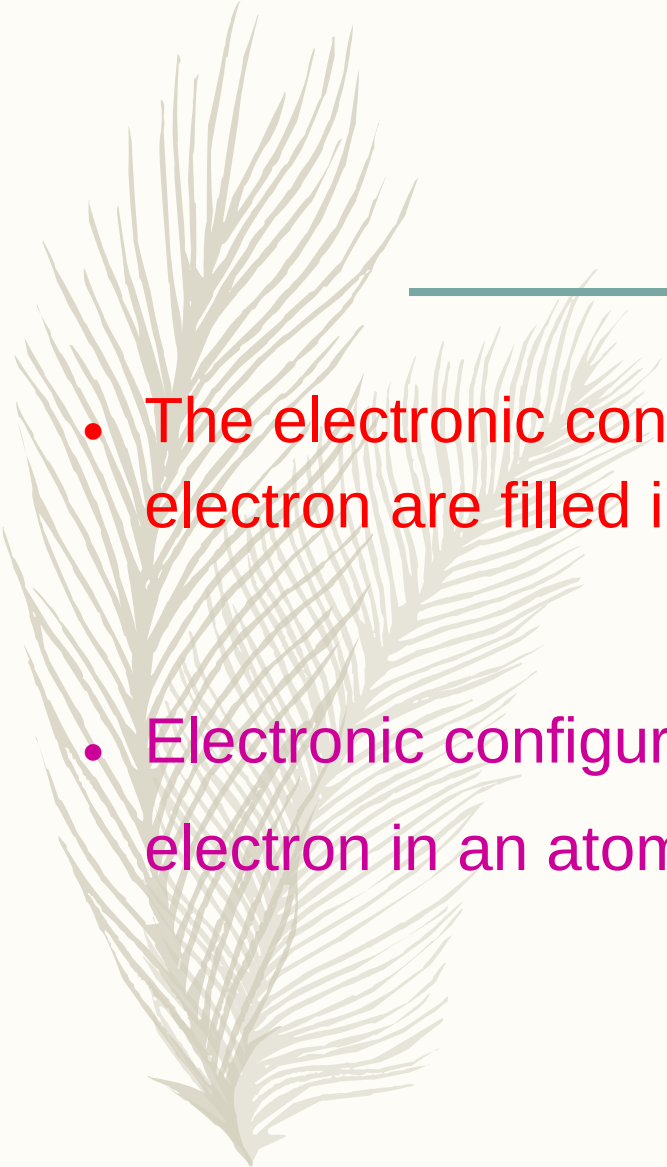


- The value of ℓ indicates s, p, or d subshell.
- Each black line represents one orbital (value count of m_ℓ .)
- The direction of the arrows represent m_s .

Number	Symbol	Possible Values
Principal Quantum Number	n	1, 2, 3, 4, ...
Angular Momentum Quantum Number	ℓ	0, 1, 2, 3, ..., (n - 1)
Magnetic Quantum Number	m_ℓ	$-\ell, \dots, -1, 0, 1, \dots, \ell$
Spin Quantum Number	m_s	$+\frac{1}{2}, -\frac{1}{2}$

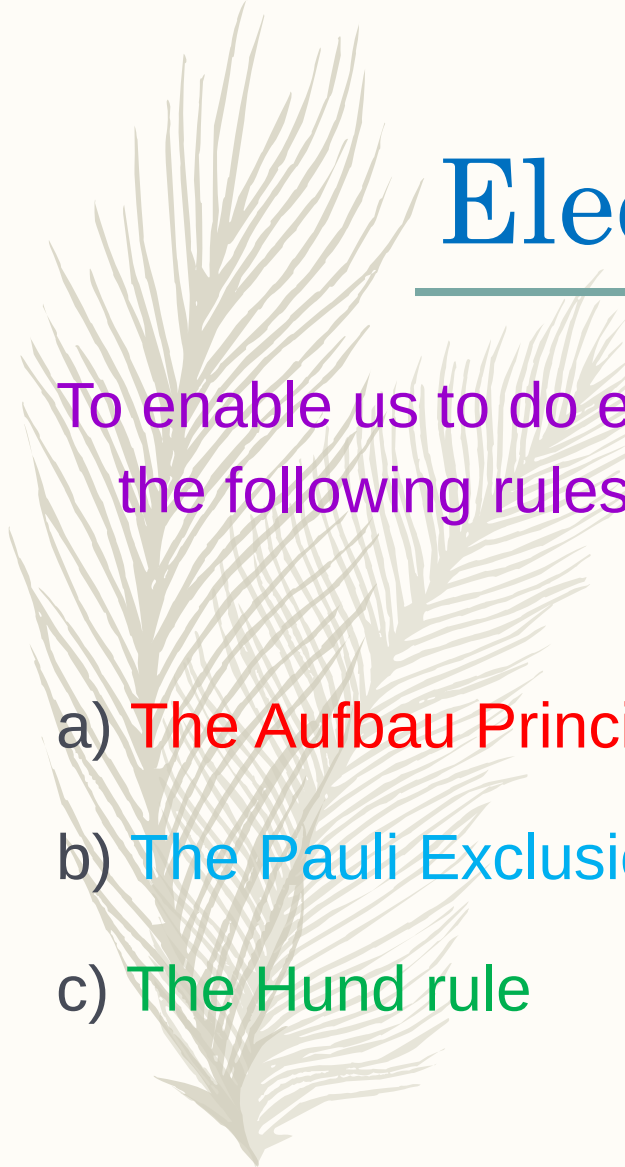
Electronic Configuration





Introduction

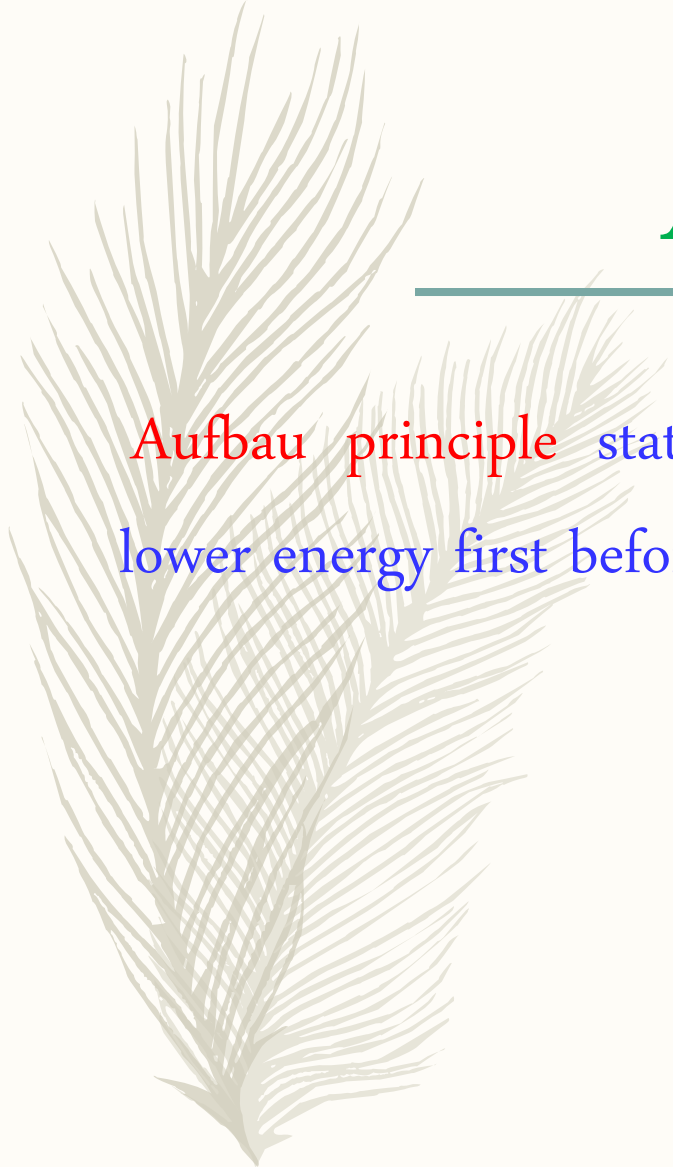
- The electronic configuration of an atom show how electron are filled in the orbital.
- Electronic configuration describes the arrangement of electron in an atom.



Electronic Configuration

To enable us to do electronic configuration, we have to obey the following rules:

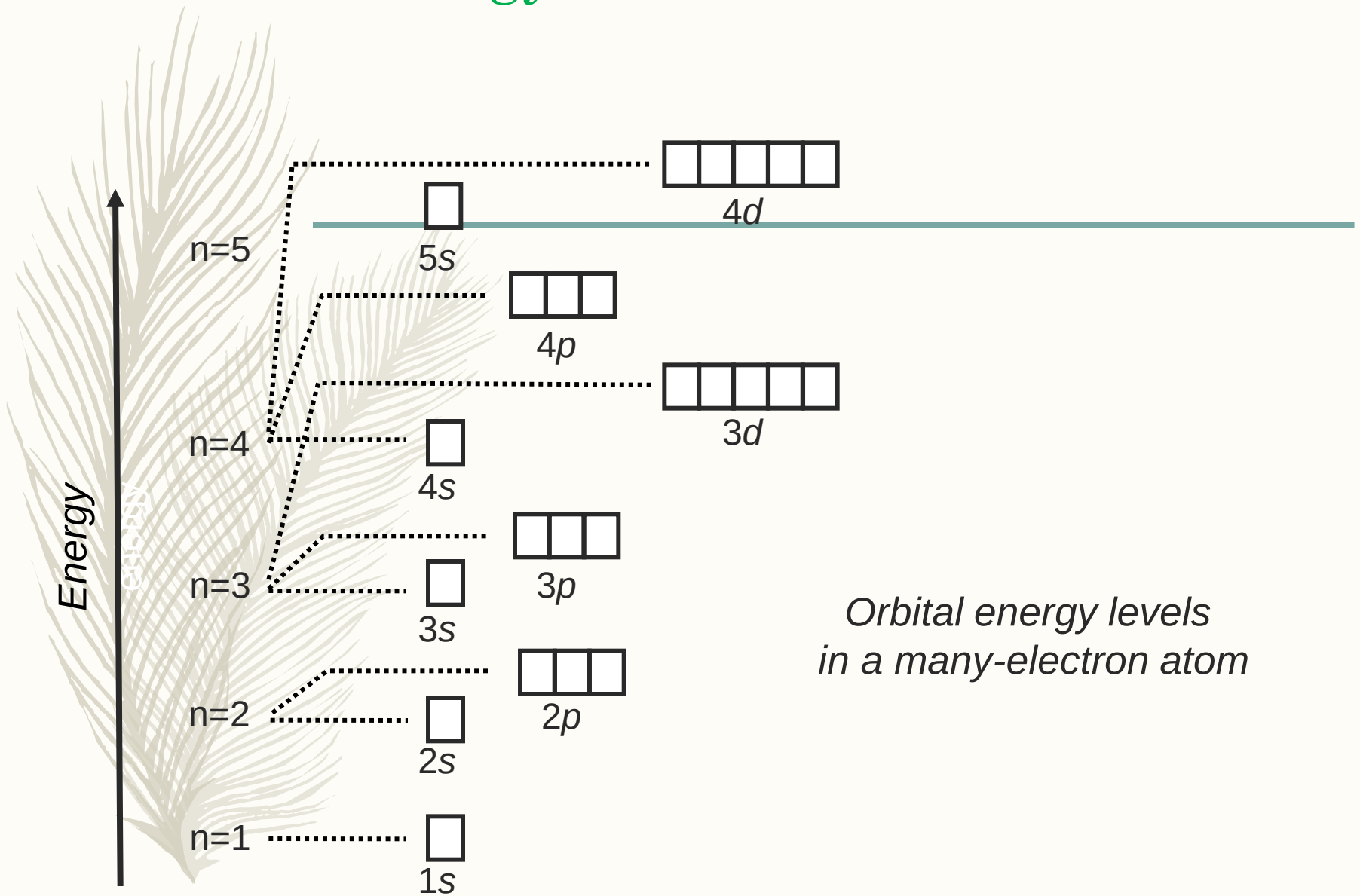
- a) The Aufbau Principle
- b) The Pauli Exclusion Principle
- c) The Hund rule



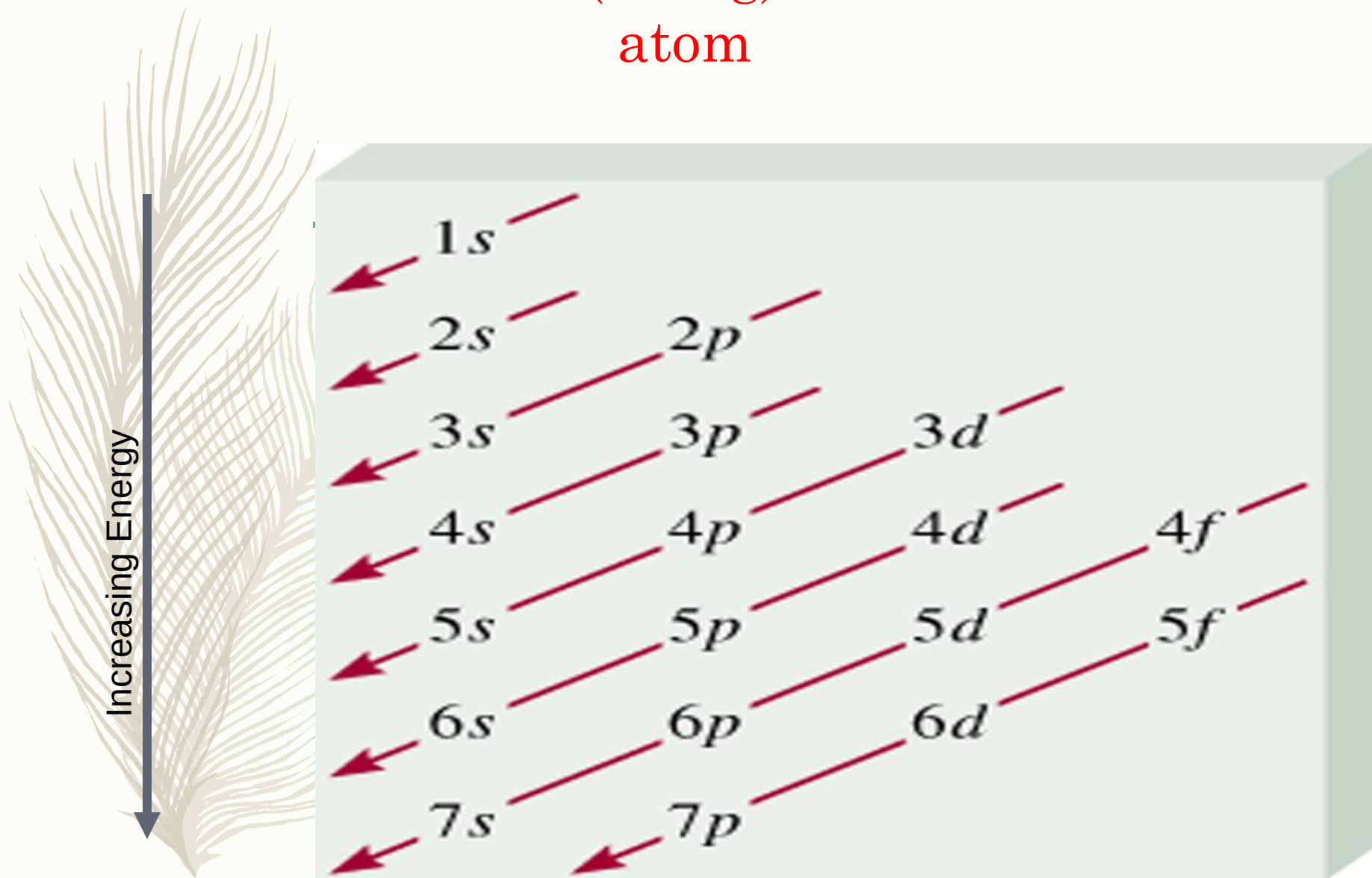
Aufbau Principle

Aufbau principle states that electron must occupy orbitals of lower energy first before orbitals of higher energy are filled

Relative Energy Level of Atomic Orbitals



Order of orbitals (filling) in multi-electron atom



Worked Example 4.2

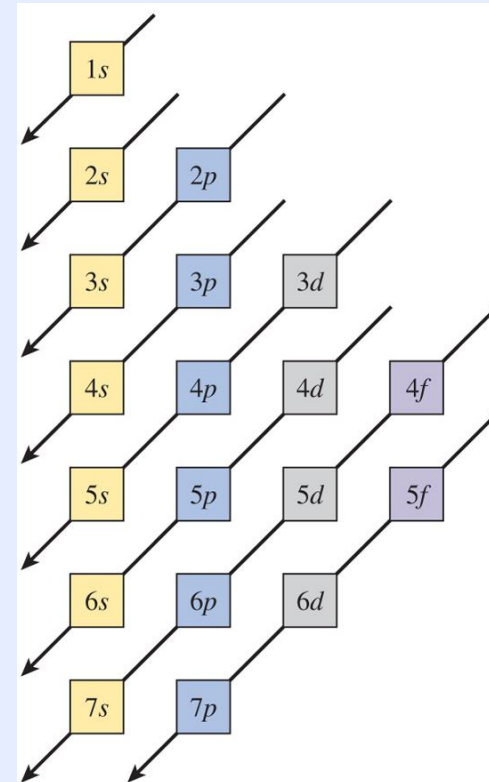
Without using a periodic table, give the ground-state electron configuration and block designation (*s*-, *p*-, *d*-, or *f*-block) of an atom with (a) 17 electrons, (b) 37 electrons, and (c) 22 electrons. Classify each atom as a main group element or transition metal.

Strategy Use the figure at right to assign electrons to orbitals in the correct order. Recall that an *s* subshell has one orbital, a *p* subshell has three orbitals, and a *d* subshell has five orbitals. Remember, too, that each orbital can accommodate a maximum of two electrons.

Solution

- (a) $1s^2 2s^2 2p^6 3s^2 3p^5$, *p*-block, main group
- (b) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1$, *s*-block, main group
- (c) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$, *d*-block, transition metal

Think About It Consult the figure to confirm your answer.



1	1	2	THE PERIODIC TABLE										13	14	15	16	17	18
1	1	2											3	4	5	6	7	8
	H	He											B	C	N	O	F	Ne
	1.0079	4.0026											10.811	12.011	14.007	15.999	18.998	20.18
2	3	4											13	14	15	16	17	18
	Li	Be											Al	Si	P	S	Cl	Ar
	6.941	9.0122											26.982	28.086	30.97	32.06	35.453	39.948
3	11	12	3	4	5	6	7	8	9	10	11	12	31	32	33	34	35	36
	Na	Mg	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
	22.99	24.305	44.956	47.88	50.942	51.996	54.938	55.847	58.933	58.693	63.546	65.39	69.723	72.61	74.92	78.96	79.904	83.8
4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
	39.098	40.078	44.956	47.88	50.942	51.996	54.938	55.847	58.933	58.693	63.546	65.39	69.723	72.61	74.92	78.96	79.904	83.8
5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
	85.468	87.62	88.906	91.224	92.906	95.94	(98.91)	101.07	102.91	106.42	107.87	112.41	114.82	118.71	121.76	127.6	126.9	131.29
6	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
	Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
	132.91	137.33	138.91	178.49	180.95	183.84	186.21	190.23	192.22	195.08	196.97	200.59	204.38	207.2	208.98	(209)	(210)	(222)
7	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104
	Fr	Ra	Ac	Rf	Ha	Sg	Ns	Hs	Mt	Unn	Unu							
	(223)	(226)	(227)	(261.1)	(262.1)	(263.1)	(262.1)	(265.1)	(266.1)	(268)	(269)							

Lanthanide Series	58	59	60	61	62	63	64	65	66	67	68	69	70	71
	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
	140.12	140.91	144.24	(144.9)	150.36	151.97	157.25	158.93	162.5	164.93	167.26	168.93	173.04	174.97
Actinide Series	90	91	92	93	94	95	96	97	98	99	100	101	102	103
	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
	232.04	231.04	238.03	(237)	(244.1)	(243.1)	(247.1)	(247.1)	(251.1)	(252.1)	(257.1)	(258.1)	(259.1)	(262.1)

Order of orbitals (filling) in multi-electron atom

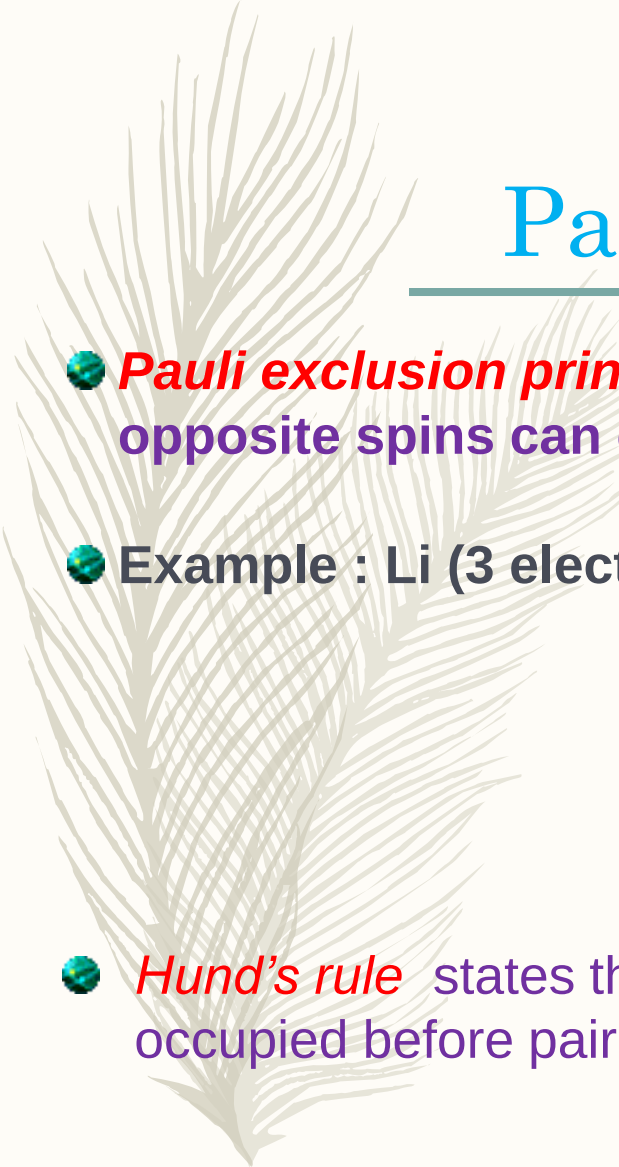
- The order of filling orbitals is:



- Start with the 1s orbital and move downward, following the arrows.
- Example:

(a) ${}_4\text{Be}$ Electronic configuration ${}_4\text{Be} : 1s^2 2s^2$

(b) ${}_{10}\text{Ne}$ Electronic configuration ${}_{10}\text{Ne} : 1s^2 2s^2 2p^6$



Pauli Exclusion Principle

- ***Pauli exclusion principle*** states that only two electrons of opposite spins can occupy each orbital.
- Example : Li (3 electrons)

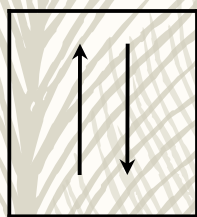
Hund's Rule

- ***Hund's rule*** states that each orbital in a subshell must be singly occupied before pair up.

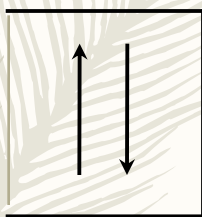
Electronic Configuration

Method 1: Orbital diagram

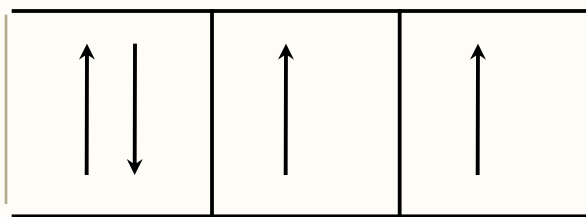
Example: ${}_8\text{O}$



1s



2s



2p

Box

Electronic Configuration

Method 2: s,p,d,f notation

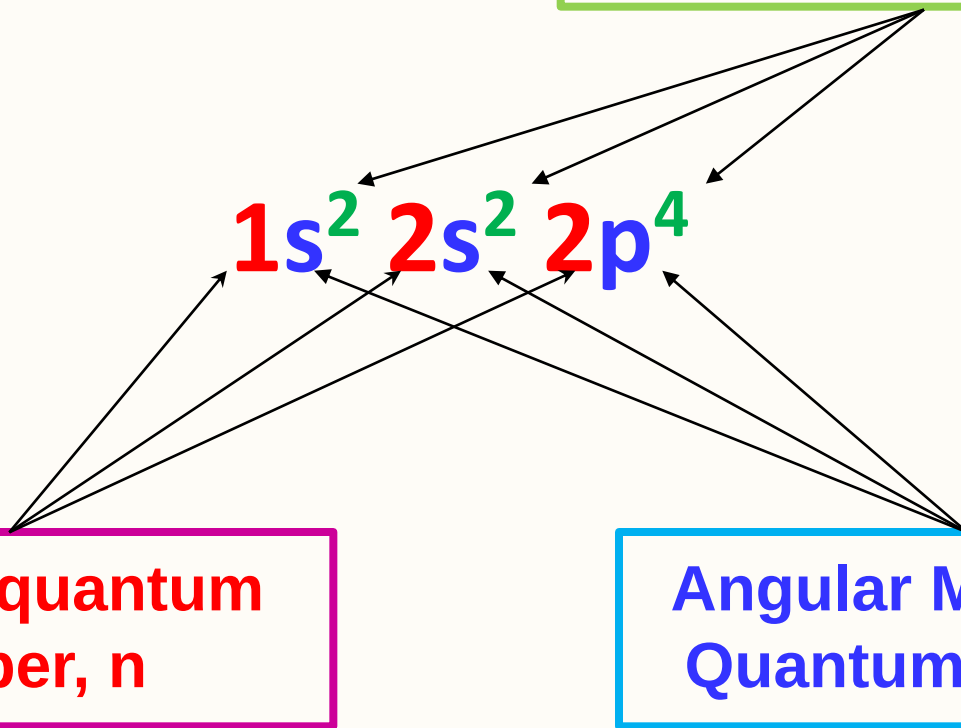
Example: ${}_8\text{O}$

Number of electrons
in the subshells

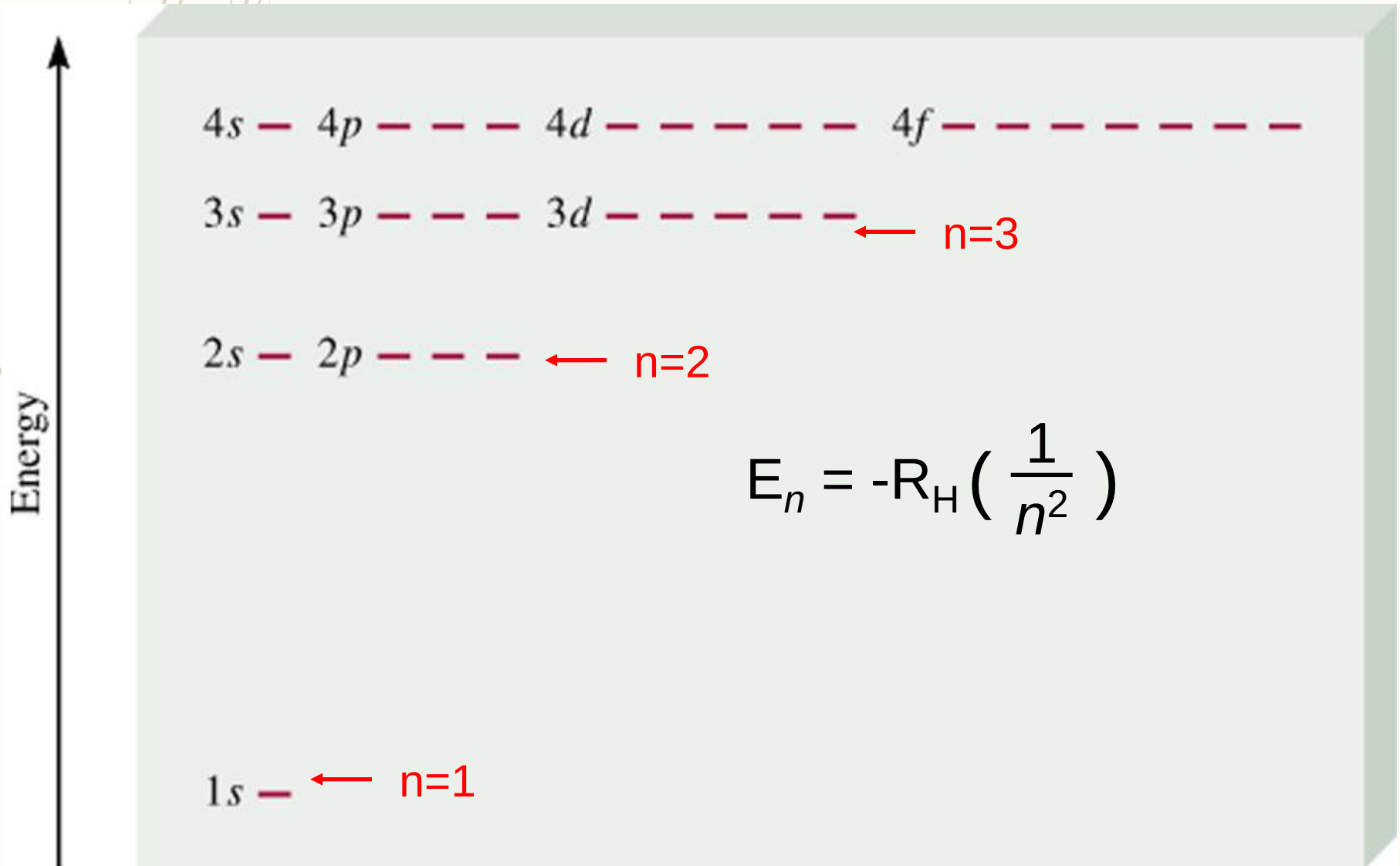
1s² **2**s² **2**p⁴

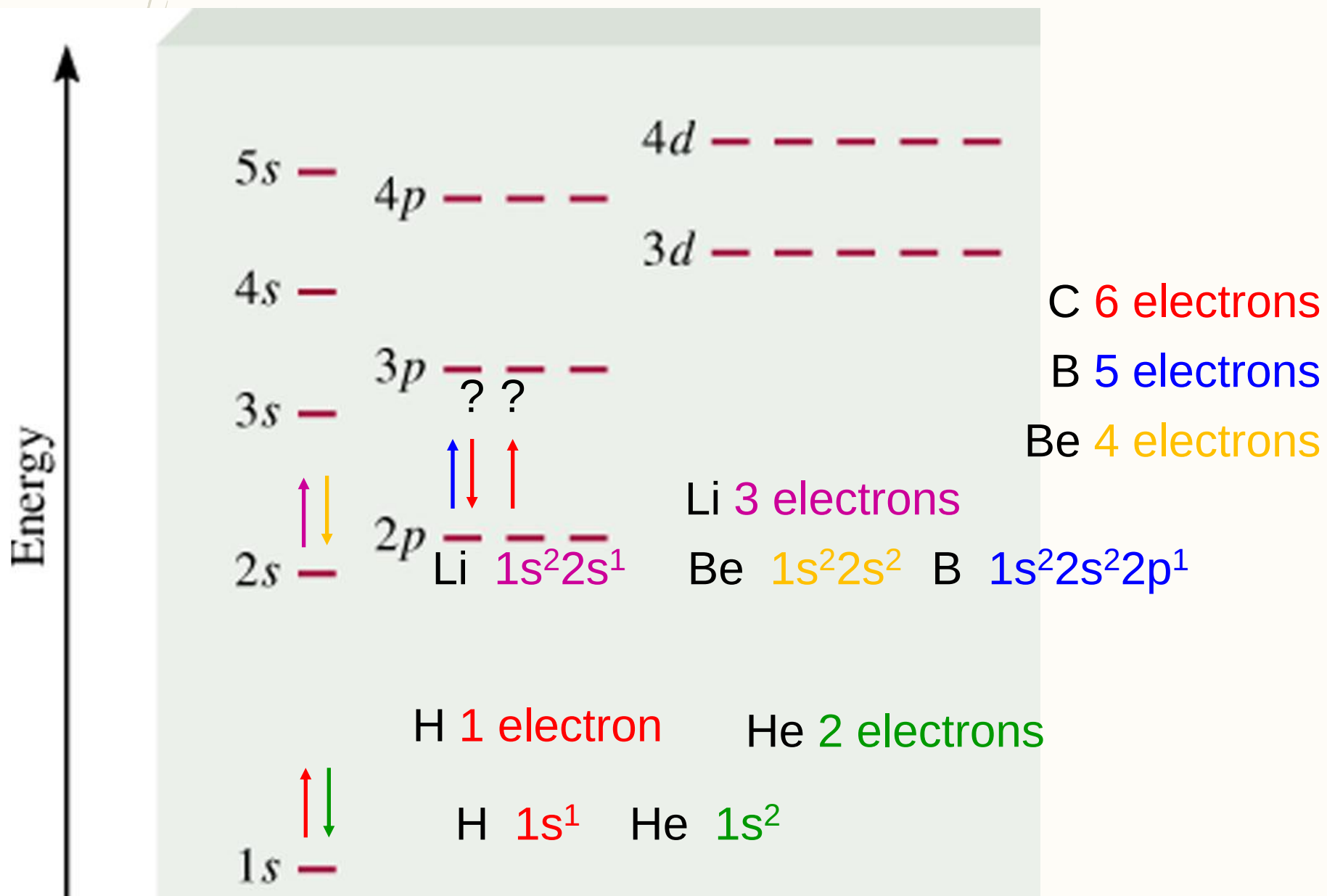
Principal quantum
number, n

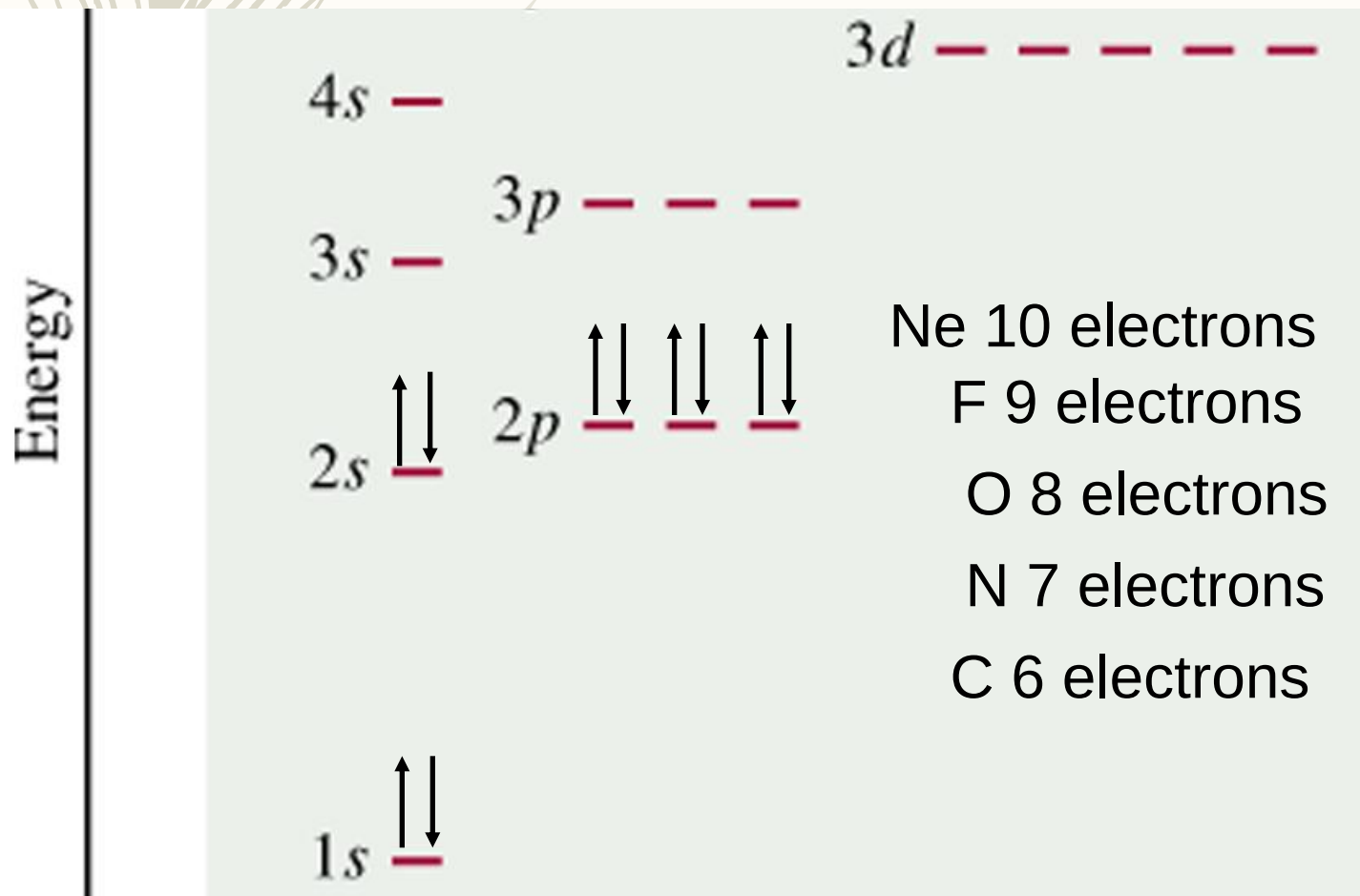
Angular Momentum
Quantum Number, l



Energy only depends on principal quantum number ***n***







Ne 10 electrons

F 9 electrons

O 8 electrons

N 7 electrons

C 6 electrons

Ne $1s^2 2s^2 2p^6$

F $1s^2 2s^2 2p^5$

O $1s^2 2s^2 2p^4$

N $1s^2 2s^2 2p^3$

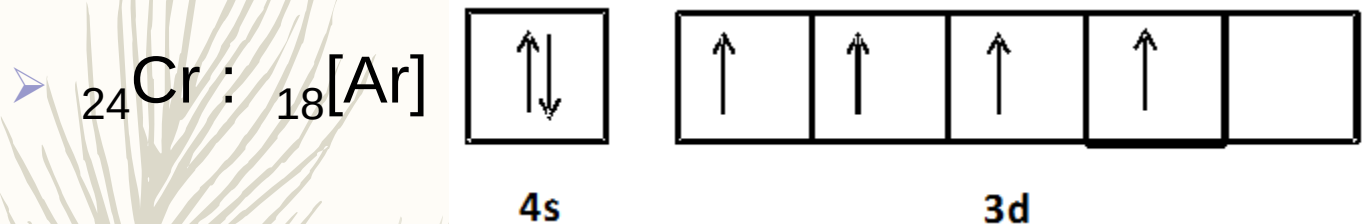
C $1s^2 2s^2 2p^2$

The Anomalous Electronic Configurations of Cr and Cu

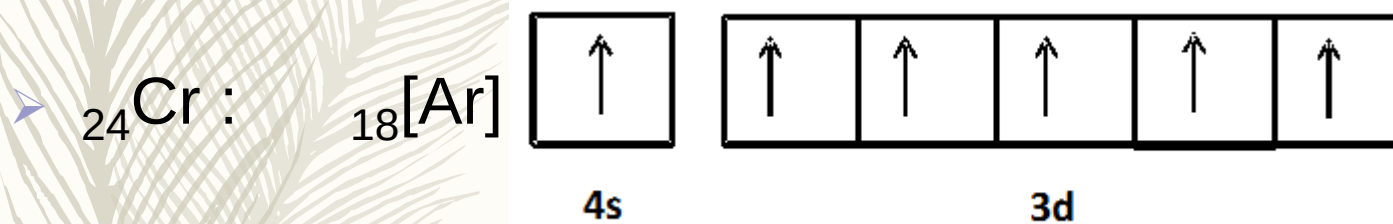
 Cr and Cu have electron configurations which are inconsistent with the Aufbau principle. The anomalous are explained on the basis that a filled or half-filled orbital is more stable.

Element	Expected (Aufbau Principle)	Observed/actual
Cr (Z=24)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
Cu (Z=29)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

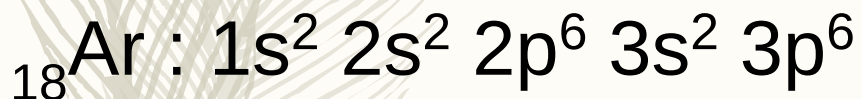
Orbital diagram



3d orbital with a half filled orbital arrangement are more stable.



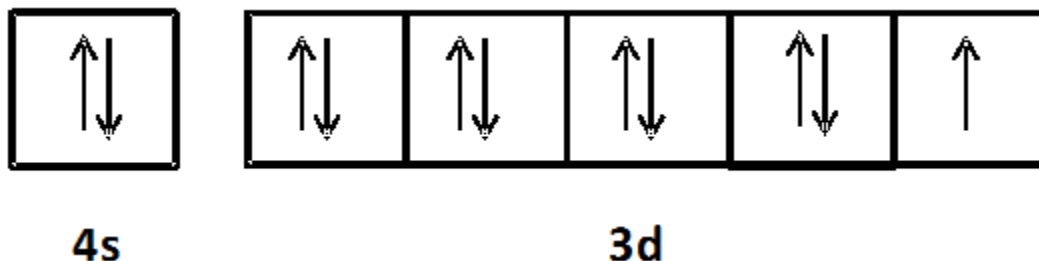
Actual



*Half filled orbital arrangement increase stability of Cr atom

Copper expected orbital notation (Aufbau Principle)

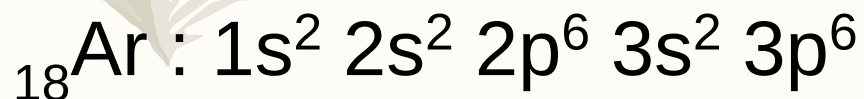
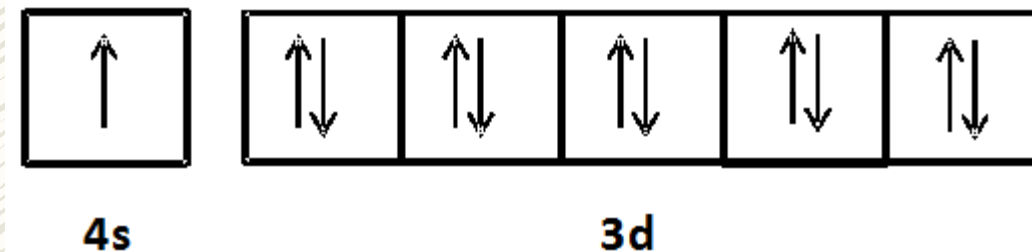
➤ Cu : $_{18}[\text{Ar}]$



3d orbital with fully filled orbital arrangement is more stable.

Copper actual orbital notation

➤ Cu : $_{18}[\text{Ar}]$





What is the electron configuration of Mg?

Mg 12 electrons

$1s < 2s < 2p < 3s < 3p < 4s$

$1s^2 2s^2 2p^6 3s^2$ $2 + 2 + 6 + 2 = 12$ electrons

Abbreviated as $[\text{Ne}]3s^2$ $[\text{Ne}] 1s^2 2s^2 2p^6$



What are the possible quantum numbers for the last (outermost) electron in Cl?

Cl 17 electrons $1s < 2s < 2p < 3s < 3p < 4s$

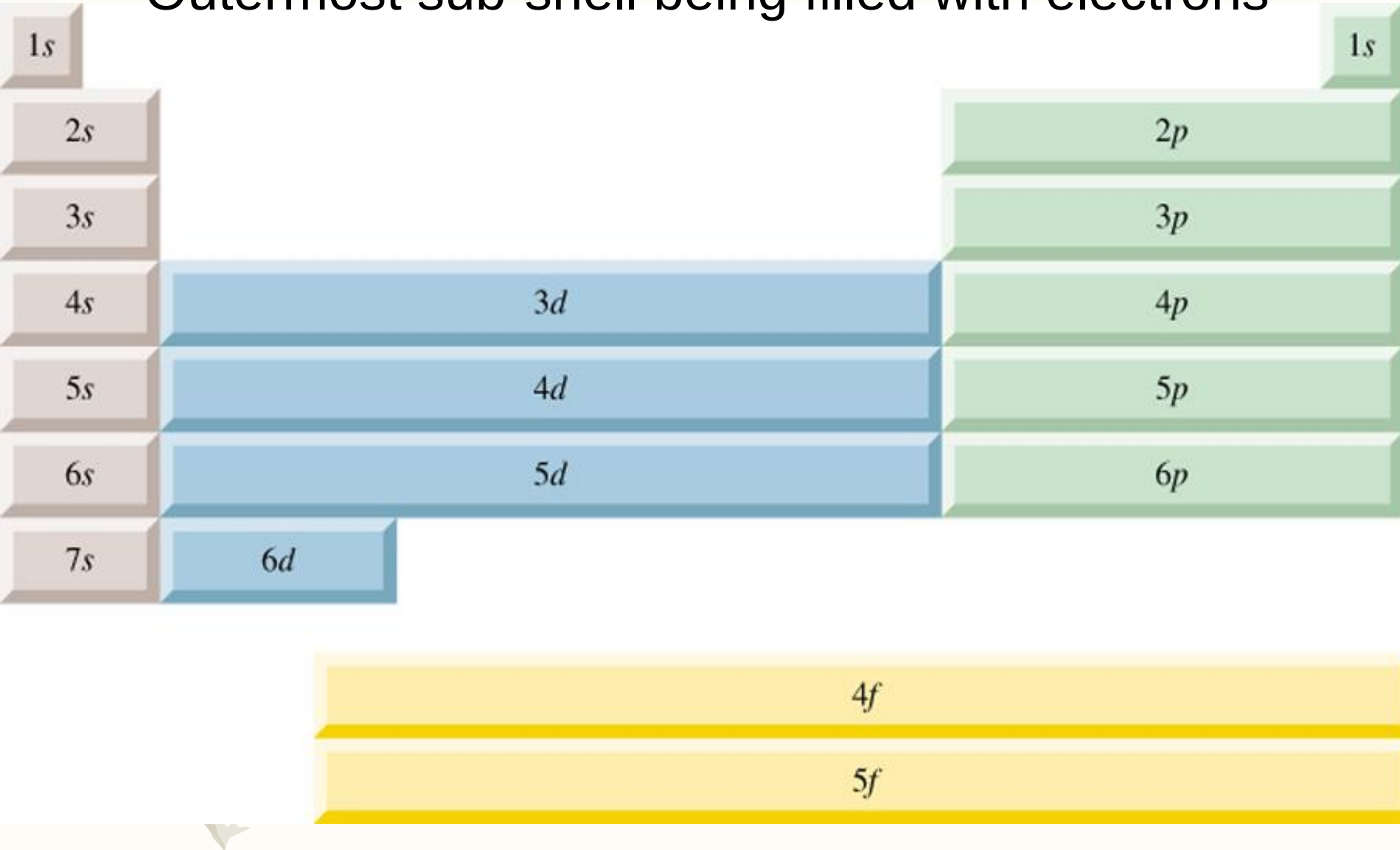
$1s^2 2s^2 2p^6 3s^2 3p^5$ $2 + 2 + 6 + 2 + 5 = 17$ electrons

Last electron added to 3p orbital

$n = 3$ $l = 1$ $m_l = -1, 0, \text{ or } +1$ $m_s = \frac{1}{2} \text{ or } -\frac{1}{2}$

Periodic table and electron configuration.

Outermost sub-shell being filled with electrons



Electron configuration for ions

- When atoms lose or gain electrons they achieve noble gas e^- structure.
- Metals lose valence e^- 's. e.g. $\text{Na} \rightarrow \text{Na}^+$
- Non-metals gain e^- 's. e.g. $\text{Cl} \rightarrow \text{Cl}^-$
- Transition elements always lose their 4s e^- 's first and they can lose more than 2 e^- 's

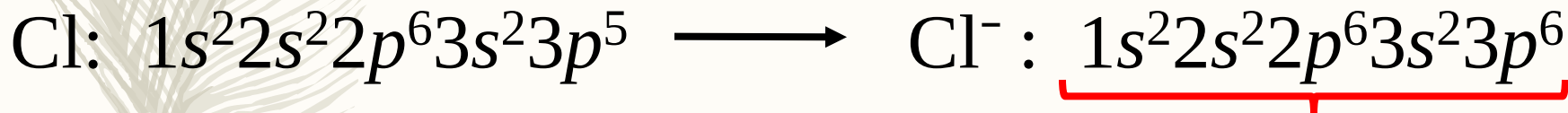
Electron Configurations of Ions

To write the electron configuration of an ion formed by a main group element:

- 1) Write the configuration for the atom.
- 2) Add or remove the appropriate number of electrons.



10 electrons total,
isoelectronic with Ne



18 electrons total,
isoelectronic with Ar

Worked Example

Write electron configurations for the following ions of main group elements:

(a) N^{3-} , (b) Ba^{2+} , and (c) Be^{2+} .

Strategy First write electron configurations for the atoms. Then add electrons (for anions) and remove electrons (for cations) to account for the charge.

Solution

(a) $[\text{He}]2s^22p^6$ or $[\text{Ne}]$

(b) $[\text{Kr}]5s^24d^{10}5p^6$ or $[\text{Xe}]$

(c) $1s^2$ or $[\text{He}]$

Think About It Be sure to add electrons to form an anion, and remove electrons to form a cation.

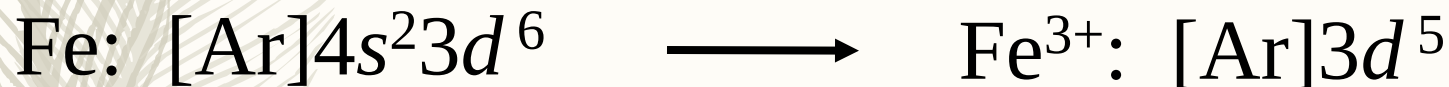
Ions of *d*-Block Elements

Ions of *d*-block elements are formed by removing electrons first from the shell with the *highest* value of *n*.

For Fe to form Fe^{2+} , two electrons are lost from the 4s subshell not the 3d.



Fe can also form Fe^{3+} , in which case the third electron is removed from the 3d subshell.



Worked Example

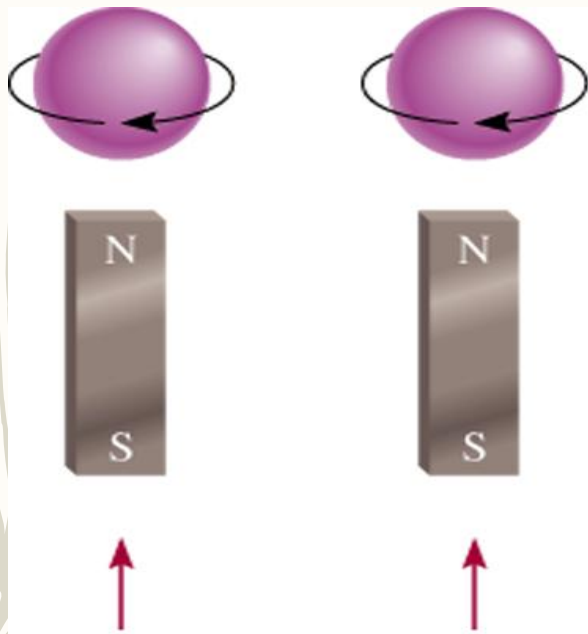
Write electron configurations for the following ions of *d*-block elements: (a) Zn^{2+} , (b) Mn^{2+} , and (c) Cr^{3+} .

Strategy First write electron configurations for the atoms. Then add electrons (for anions) and remove electrons (for cations) to account for the charge. The electrons removed from a *d*-block element must come first from the outermost *s* subshell, not the partially filled *d* subshell.

Solution



Think About It Be sure to *add* electrons to form an anion, and *remove* electrons to form a cation. Also, double-check to make sure that electrons removed from a *d*-block element come first from the *ns* subshell and then, if necessary, from the $(n - 1)d$ subshell.



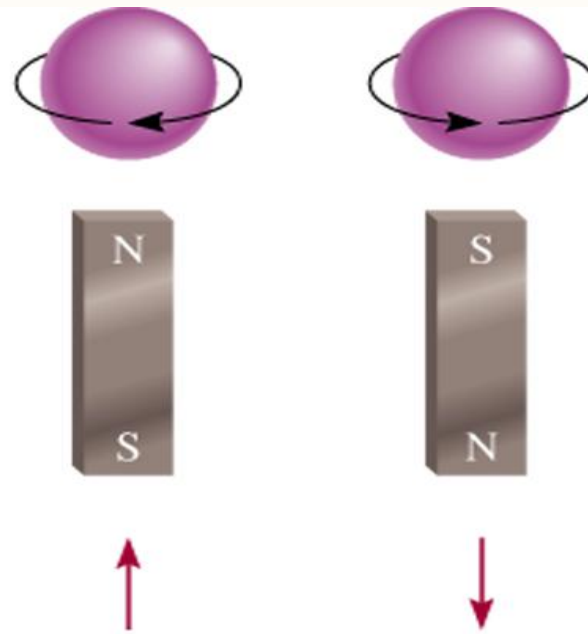
Paramagnetic

Paramagnetic substances are those that contain net unpaired spins and are attracted by a magnet.

unpaired electrons



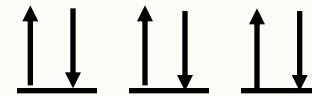
2p



Diamagnetic

Diamagnetic substances do not contain net unpaired spins and are slightly repelled by a magnet.

all electrons paired



2p